



Introduction

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Hight-Weight

Obesity

Computing 'g'

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Least square

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Prediction

Regression Analysis What and Why ?

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Example: Hight-Weight chart

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What is Considered the Right Weight for My Height?

**The table below has been updated to show both Metric and Imperial measurements i.e. Inches/Centimeters - Pounds/Kilograms.*

Adults Weight to Height Ratio Chart		
Height	Female	Male
4' 6" (137 cm)	63/77 lb (28.5/34.9 kg)	63/77 lb (28.5/34.9 kg)
4' 7" (140 cm)	68/83 lb (30.8/37.6 kg)	68/84 lb (30.8/38.1 kg)
4' 8" (142 cm)	72/88 lb (32.6/39.9 kg)	74/90 lb (33.5/40.8 kg)
4' 9" (145 cm)	77/94 lb (34.9/42.6 kg)	79/97 lb (35.8/43.9 kg)
4' 10" (147 cm)	81/99 lb (36.4/44.9 kg)	85/103 lb (38.5/46.7 kg)
4' 11" (150 cm)	86/105 lb (39/47.6 kg)	90/110 lb (40.8/49.9 kg)
5' 0" (152 cm)	90/110 lb (40.8/49.9 kg)	95/117 lb (43.1/53 kg)
5' 1" (155 cm)	95/116 lb (43.1/52.6 kg)	101/123 lb (45.8/55.8 kg)
5' 2" (157 cm)	99/121 lb (44.9/54.9 kg)	106/130 lb (48.1/58.9 kg)
5' 3" (160 cm)	104/127 lb (47.2/57.6 kg)	112/136 lb (50.8/61.6 kg)
5' 4" (163 cm)	108/132 lb (49/59.9 kg)	117/143 lb (53/64.8 kg)



Example: Hight-Weight chart

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Adult Male and Female Height to Weight Ratio Chart ¹

Author: Disabled World : Contact: www.disabled-world.com

Published: 2017-11-30 : (Rev. 2020-03-05)

¹Ref: <https://www.disabled-world.com/calculators-charts/height-weight.php>



Weight-Hight regression

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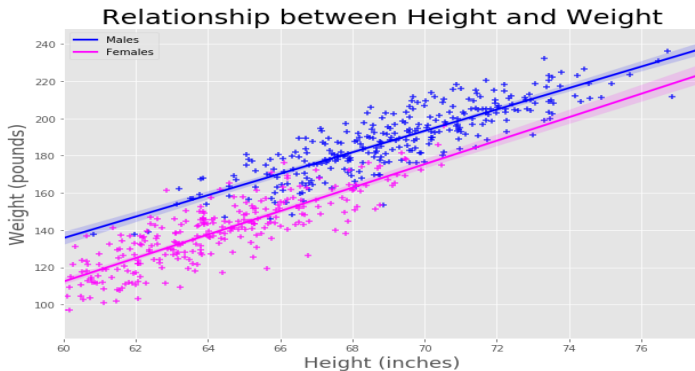


Figure: Weight vs Hight



Example: Obesity

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- Worldwide, at least 2.8 million people die each year as a result of being overweight or obese, and an estimated 35.8 million (2.3%) of global DALYs are caused by overweight or obesity.²
- What are obesity and overweight ?
Overweight and obesity are defined as abnormal or excessive fat accumulation that may impair health.
- For adults, WHO defines overweight and obesity as follows:
 - overweight is a BMI greater than or equal to 25; and
 - obesity is a BMI greater than or equal to 30.
- Body mass index (BMI) is a simple index of weight-for-height that is commonly used to classify overweight and obesity in adults. It is defined as a person's weight in kilograms divided by the square of his height in meters (kg/m^2).

² Ref: <https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight>



Example: Obesity chart for girls (5-19yr)

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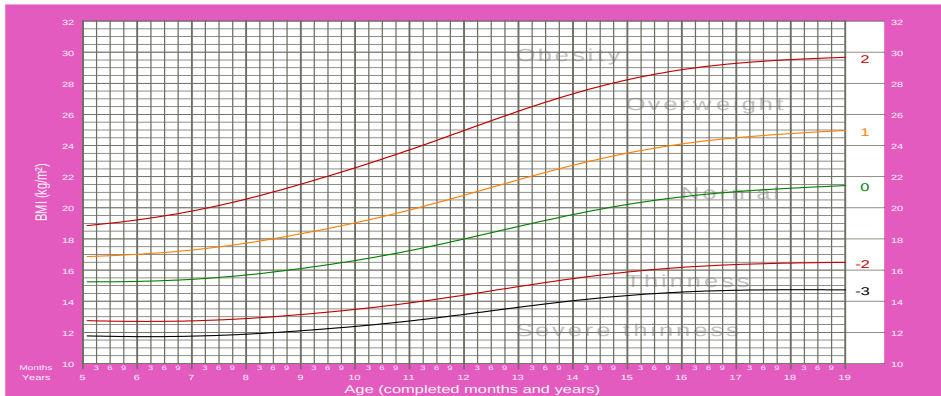
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BMI-for-age GIRLS

5 to 19 years (z-scores)





Example: Obesity chart for boys (5-19yrs)

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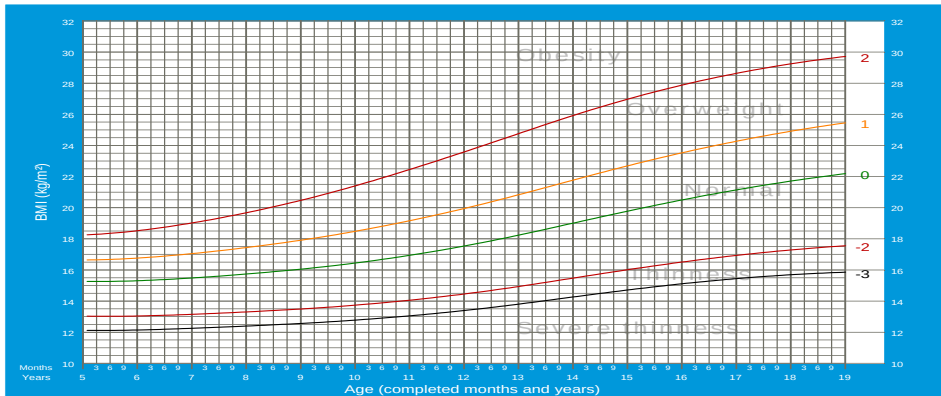
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BMI-for-age BOYS

5 to 19 years (z-scores)





What is the value of 'g' ?

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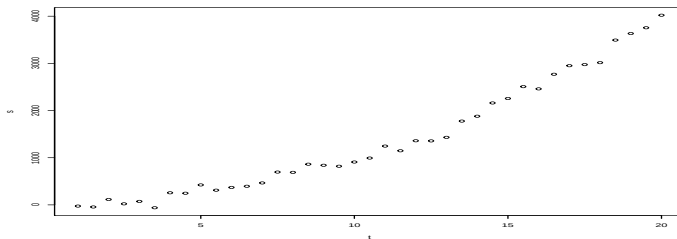


Figure: Free fall

$$S = ut + \frac{1}{2}gt^2$$



Two treatment comparison

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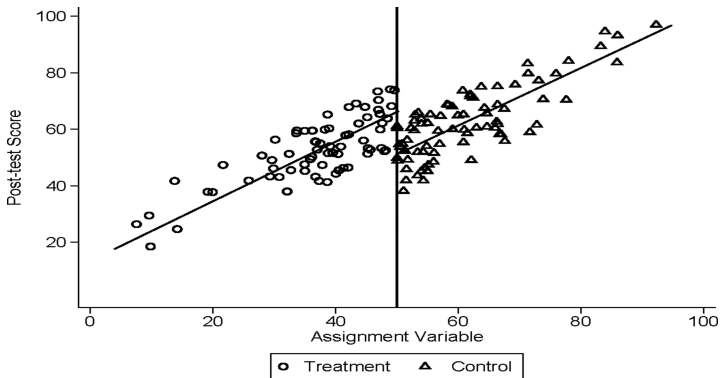


Figure: Linear Treatment effect model



Why regression?

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- Regression is a very natural attempt to answer many queries that come in human mind and scientific work.
- The information we gather about a natural phenomena or a controlled experiment are often incomplete.
- Regression is one of the ways to make these information complete based on the available data.
- In other words, it an attempt to access beyond than that has been already observed.



What is regression?

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Definition

Let $(Y, \mathbf{X})$ be a random vector. The conditional expectation of Y given $\mathbf{X} = \mathbf{x}$, is known as the regression of Y on $\mathbf{X}$. It can be denoted as

$$\hat{y} = g(\mathbf{x}, \beta) = E(Y|\mathbf{X} = \mathbf{x})$$

- $g(\mathbf{x}, \beta)$ can be a line, curve, plane, surface etc. or may be unknown
- $\mathbf{x}$ can be stochastic or non-stochastic
- Y is always stochastic or a random viable



Linear and non-linear regression

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Consider a data set $D = \{(\mathbf{x}_i, y_i) | \mathbf{x}_i \in \mathbb{R}^k, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$ where x_i s are non-stochastic but y_i are stochastic and realized values of random variable Y_i s respectively.

Definition

If the relation, $g(\mathbf{x}, \beta)$, between the **response variable** y and the **regressor variable** $\mathbf{x}$ is linear in parameter (β) then it is called a **linear regression**.

e.g. **Linear regression:**

$$y = \beta_0 + \beta_1 x + \epsilon$$

$$y = \beta_0 + \beta_1 e^x + \epsilon$$

e.g. **Non-Linear regression:**

$$y = \frac{1}{\beta_0 + \beta_1 x} + \epsilon$$

$$y = \beta_0 \cos(\beta_1 + \beta_2 x) + \epsilon$$

where, ϵ is random error.



Some varieties of regression

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- Linear Model (LM):
 - Simple linear regression
 - Multiple linear regression
 - Polynomial regression
- Generalized linear model (GLM)
 - Logit-model
 - Probit-model
 - Poisson-regression
- Isotonic regression
- Spline regression etc.



Simple linear regression

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- Consider a data set $D = \{(x_i, y_i) | x_i \in \mathbb{R}, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$
- x_i s are non stochastic
- y_i s are stochastic and realized values of random variable Y_i s

Problem statement

We are interested to have a prediction line

$$\hat{y} = g(x, \beta_0, \beta_1) = \beta_0 + \beta_1 x$$

which will approximate well the y values if the x values are known.



Simple linear regression : Example

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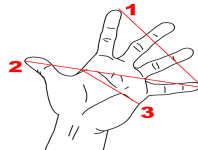
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Y= Hight



X= length of palm of hand as shown in 2

Figure: Palm length vs Hight

Can we have a prediction line $\hat{y} = \beta_0 + \beta_1 x$ which will approximate well the hight of a person if his/her palm length(2) is known ?



Least square estimate

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- What do we mean by “approximate well”?

ANS: Minimum distance between the predicted ($\hat{y}_i$'s) and the true (y_i 's) values of Y .

- What will the notion of distance ?

ANS: There could be many. But we will consider either absolute or square/ Euclidean distance.

- Given the data how can we obtain the values of β_0 and β_1 ?

ANS: We will consider such values of β_0 and β_1 that will minimize the square/ Euclidean distance between the predicted ($\hat{y}_i$'s) and the true (y_i 's) values of Y .

This is known as the least squared method of estimation



Least square estimate

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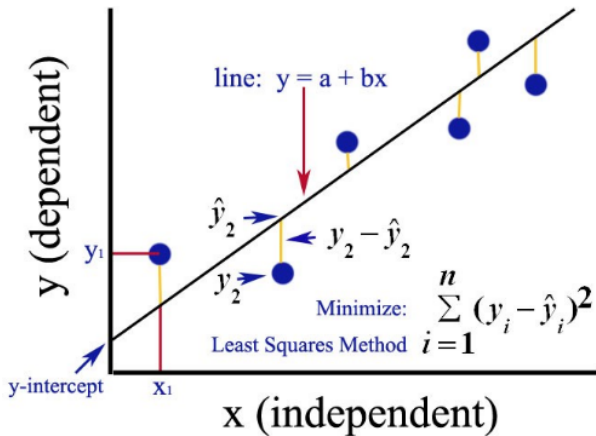


Figure: Least square



Parameter estimation

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Least Squared condition

The least squared condition to estimate the model parameters is to minimize

$$S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2. \quad (1)$$

with respect to β_0 and β_1 .

If $(\hat{\beta}_0, \hat{\beta}_1)$ minimizes $S(\beta_0, \beta_1)$ then their values can be obtained by solving the **normal equations**

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_0} = 0 \quad \implies \quad n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i = \sum_i y_i \quad (2)$$

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_1} = 0 \quad \implies \quad \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2 = \sum_i y_i x_i \quad (3)$$



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Estimated parameters

Defining $S_{xy} = \sum_i (y_i - \bar{y})(x_i - \bar{x})$ we have the solutions as

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Least squared prediction line

For any x such as old x_i s or some x_{new} the prediction line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$



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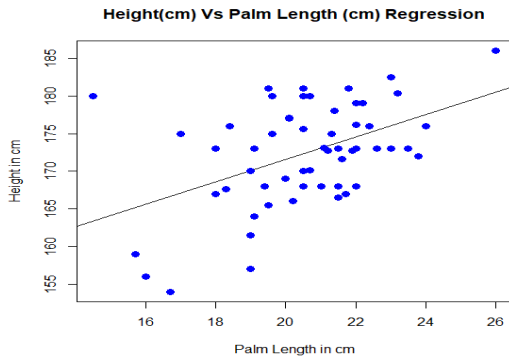
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$$\hat{\beta}_0 = 141.8916$$

$$\hat{\beta}_1 = 1.4833$$



What more?

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- Can we have a prediction interval for $\hat{y}$?
- Can we test for $H_0 : \beta_0 = 140$ vs $H_1 : \beta_0 > 140$?
- Can we test for $H_0 : \beta_1 = 1.5$ vs $H_1 : \beta_1 \neq 1.5$?
- What will be the distribution of estimated error ?
- If we incorporate more regressor variables then how significantly the error can be reduced ?

We are not yet ready to answer these important questions!!!!
Results from Linear Algebra and Multivariate Analysis can help us.



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Matrix Differentiation

(and some other stuff)

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1 Introduction

Throughout this presentation I have chosen to use a *symbolic matrix notation*. This choice was not made lightly. I am a strong advocate of index notation, when appropriate. For example, index notation greatly simplifies the presentation and manipulation of differential geometry. As a rule-of-thumb, if your work is going to primarily involve differentiation with respect to the spatial coordinates, then index notation is almost surely the appropriate choice.

In the present case, however, I will be manipulating large systems of equations in which the matrix calculus is relatively simply while the matrix algebra and matrix arithmetic is messy and more involved. Thus, I have chosen to use symbolic notation.

2 Notation and Nomenclature

Definition 1 Let $\mathbf{a}_{ij} \in \mathfrak{R}$, $i = 1, 2, \dots, \mathbf{m}$, $j = 1, 2, \dots, \mathbf{n}$. Then the ordered rectangular array

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_{11} & \mathbf{a}_{12} & \cdots & \mathbf{a}_{1\mathbf{n}} \\ \mathbf{a}_{21} & \mathbf{a}_{22} & \cdots & \mathbf{a}_{2\mathbf{n}} \\ \vdots & \vdots & & \vdots \\ \mathbf{a}_{\mathbf{m}1} & \mathbf{a}_{\mathbf{m}2} & \cdots & \mathbf{a}_{\mathbf{m}\mathbf{n}} \end{bmatrix} \quad (1)$$

is said to be a real *matrix* of dimension $\mathbf{m} \times \mathbf{n}$.

When writing a matrix I will occasionally write down its typical element as well as its dimension. Thus,

$$\mathbf{A} = [\mathbf{a}_{ij}], \quad i = 1, 2, \dots, \mathbf{m}; j = 1, 2, \dots, \mathbf{n}, \quad (2)$$

denotes a matrix with $\mathbf{m}$ rows and $\mathbf{n}$ columns, whose typical element is $\mathbf{a}_{ij}$. Note, the first subscript locates the *row* in which the typical element lies while the second subscript locates the *column*. For example, $\mathbf{a}_{jk}$ denotes the element lying in the j th row and k th column of the matrix $\mathbf{A}$.

Definition 2 A *vector* is a matrix with only one column. Thus, all vectors are inherently column vectors.

Convention 1

Multi-column matrices are denoted by boldface uppercase letters: for example, $\mathbf{A}, \mathbf{B}, \mathbf{X}$. Vectors (single-column matrices) are denoted by boldfaced lowercase letters: for example, $\mathbf{a}, \mathbf{b}, \mathbf{x}$. I will attempt to use letters from the beginning of the alphabet to designate known matrices, and letters from the end of the alphabet for unknown or variable matrices.

Convention 2

When it is useful to explicitly attach the matrix dimensions to the symbolic notation, I will use an underscript. For example, $\mathbf{A}_{m \times n}$, indicates a known, multi-column matrix with m rows and n columns.

A superscript T denotes the matrix transpose operation; for example, $\mathbf{A}^T$ denotes the transpose of $\mathbf{A}$. Similarly, if $\mathbf{A}$ has an inverse it will be denoted by $\mathbf{A}^{-1}$. The determinant of $\mathbf{A}$ will be denoted by either $|\mathbf{A}|$ or $\det(\mathbf{A})$. Similarly, the rank of a matrix $\mathbf{A}$ is denoted by $\text{rank}(\mathbf{A})$. An identity matrix will be denoted by $\mathbf{I}$, and $\mathbf{0}$ will denote a null matrix.

3 Matrix Multiplication

Definition 3 Let $\mathbf{A}$ be $m \times n$, and $\mathbf{B}$ be $n \times p$, and let the product $\mathbf{AB}$ be

$$\mathbf{C} = \mathbf{AB} \quad (3)$$

then $\mathbf{C}$ is a $m \times p$ matrix, with element (i, j) given by

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} \quad (4)$$

for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, p$.

Proposition 1 Let $\mathbf{A}$ be $m \times n$, and $\mathbf{x}$ be $n \times 1$, then the typical element of the product

$$\mathbf{z} = \mathbf{Ax} \quad (5)$$

is given by

$$z_i = \sum_{k=1}^n a_{ik} x_k \quad (6)$$

for all $i = 1, 2, \dots, m$. Similarly, let $\mathbf{y}$ be $m \times 1$, then the typical element of the product

$$\mathbf{z}^T = \mathbf{y}^T \mathbf{A} \quad (7)$$

is given by

$$z_i = \sum_{k=1}^n a_{ki} y_k \quad (8)$$

for all $i = 1, 2, \dots, n$. Finally, the scalar resulting from the product

$$\alpha = \mathbf{y}^T \mathbf{Ax} \quad (9)$$

is given by

$$\alpha = \sum_{j=1}^m \sum_{k=1}^n a_{jk} y_j x_k \quad (10)$$

Proof: These are merely direct applications of Definition 3. q.e.d.

Proposition 2 Let $\mathbf{A}$ be $m \times n$, and $\mathbf{B}$ be $n \times p$, and let the product $\mathbf{AB}$ be

$$\mathbf{C} = \mathbf{AB} \quad (11)$$

then

$$\mathbf{C}^T = \mathbf{B}^T \mathbf{A}^T \quad (12)$$

Proof: The typical element of $\mathbf{C}$ is given by

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} \quad (13)$$

By definition, the typical element of $\mathbf{C}^T$, say d_{ij} , is given by

$$d_{ij} = c_{ji} = \sum_{k=1}^n a_{jk} b_{ki} \quad (14)$$

Hence,

$$\mathbf{C}^T = \mathbf{B}^T \mathbf{A}^T \quad (15)$$

q.e.d.

Proposition 3 Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ and invertible matrices. Let the product $\mathbf{AB}$ be given by

$$\mathbf{C} = \mathbf{AB} \quad (16)$$

then

$$\mathbf{C}^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1} \quad (17)$$

Proof:

$$\mathbf{CB}^{-1} \mathbf{A}^{-1} = \mathbf{ABB}^{-1} \mathbf{A}^{-1} = \mathbf{I} \quad (18)$$

q.e.d.

4 Partitioned Matrices

Frequently, I will find it convenient to deal with *partitioned matrices*¹. Such a representation, and the manipulation of this representation, are two of the relative advantages of the symbolic matrix notation.

Definition 4 Let $\mathbf{A}$ be $m \times n$ and write

$$\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{D} & \mathbf{E} \end{bmatrix} \quad (19)$$

where $\mathbf{B}$ is $m_1 \times n_1$, $\mathbf{E}$ is $m_2 \times n_2$, $\mathbf{C}$ is $m_1 \times n_2$, $\mathbf{D}$ is $m_2 \times n_1$, $m_1 + m_2 = m$, and $n_1 + n_2 = n$. The above is said to be a *partition* of the matrix $\mathbf{A}$.

¹Much of the material in this section is extracted directly from Dhrymes (1978, Section 2.7).

Proposition 4 *Let $\mathbf{A}$ be a square, nonsingular matrix of order $\mathbf{m}$. Partition $\mathbf{A}$ as*

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \quad (20)$$

so that $\mathbf{A}_{11}$ is a nonsingular matrix of order $\mathbf{m}_1$, $\mathbf{A}_{22}$ is a nonsingular matrix of order $\mathbf{m}_2$, and $\mathbf{m}_1 + \mathbf{m}_2 = \mathbf{m}$. Then

$$\mathbf{A}^{-1} = \begin{bmatrix} (\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1} & -\mathbf{A}_{11}^{-1}\mathbf{A}_{12}(\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}(\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1} & (\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1} \end{bmatrix} \quad (21)$$

Proof: Direct multiplication of the proposed $\mathbf{A}^{-1}$ and $\mathbf{A}$ yields

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (22)$$

q.e.d.

5 Matrix Differentiation

In the following discussion I will differentiate matrix quantities with respect to the elements of the referenced matrices. Although no new concept is required to carry out such operations, the element-by-element calculations involve cumbersome manipulations and, thus, it is useful to derive the necessary results and have them readily available ².

Convention 3

Let

$$\mathbf{y} = \psi(\mathbf{x}), \quad (23)$$

where $\mathbf{y}$ is an $\mathbf{m}$ -element vector, and $\mathbf{x}$ is an $\mathbf{n}$ -element vector. The symbol

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \dots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \dots & \frac{\partial y_2}{\partial x_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial y_m}{\partial x_1} & \frac{\partial y_m}{\partial x_2} & \dots & \frac{\partial y_m}{\partial x_n} \end{bmatrix} \quad (24)$$

will denote the $\mathbf{m} \times \mathbf{n}$ matrix of first-order partial derivatives of the transformation from $\mathbf{x}$ to $\mathbf{y}$. Such a matrix is called the Jacobian matrix of the transformation $\psi(\cdot)$.

Notice that if $\mathbf{x}$ is actually a scalar in Convention 3 then the resulting Jacobian matrix is a $\mathbf{m} \times 1$ matrix; that is, a single column (a vector). On the other hand, if $\mathbf{y}$ is actually a scalar in Convention 3 then the resulting Jacobian matrix is a $1 \times \mathbf{n}$ matrix; that is, a single row (the transpose of a vector).

Proposition 5 *Let*

$$\mathbf{y} = \mathbf{Ax} \quad (25)$$

²Much of the material in this section is extracted directly from Dhrymes (1978, Section 4.3). The interested reader is directed to this worthy reference to find additional results.

where $\mathbf{y}$ is $m \times 1$, $\mathbf{x}$ is $n \times 1$, $\mathbf{A}$ is $m \times n$, and $\mathbf{A}$ does not depend on $\mathbf{x}$, then

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \mathbf{A} \quad (26)$$

Proof: Since the i th element of $\mathbf{y}$ is given by

$$y_i = \sum_{k=1}^n a_{ik} x_k \quad (27)$$

it follows that

$$\frac{\partial y_i}{\partial x_j} = a_{ij} \quad (28)$$

for all $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$. Hence

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \mathbf{A} \quad (29)$$

q.e.d.

Proposition 6 *Let*

$$\mathbf{y} = \mathbf{A}\mathbf{x} \quad (30)$$

where $\mathbf{y}$ is $m \times 1$, $\mathbf{x}$ is $n \times 1$, $\mathbf{A}$ is $m \times n$, and $\mathbf{A}$ does not depend on $\mathbf{x}$, as in Proposition 5. Suppose that $\mathbf{x}$ is a function of the vector $\mathbf{z}$, while $\mathbf{A}$ is independent of $\mathbf{z}$. Then

$$\frac{\partial \mathbf{y}}{\partial \mathbf{z}} = \mathbf{A} \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (31)$$

Proof: Since the i th element of $\mathbf{y}$ is given by

$$y_i = \sum_{k=1}^n a_{ik} x_k \quad (32)$$

for all $i = 1, 2, \dots, m$, it follows that

$$\frac{\partial y_i}{\partial z_j} = \sum_{k=1}^n a_{ik} \frac{\partial x_k}{\partial z_j} \quad (33)$$

but the right hand side of the above is simply element (i, j) of $\mathbf{A} \frac{\partial \mathbf{x}}{\partial \mathbf{z}}$. Hence

$$\frac{\partial \mathbf{y}}{\partial \mathbf{z}} = \frac{\partial \mathbf{y}}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial \mathbf{z}} = \mathbf{A} \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (34)$$

q.e.d.

Proposition 7 *Let the scalar α be defined by*

$$\alpha = \mathbf{y}^T \mathbf{A} \mathbf{x} \quad (35)$$

where $\mathbf{y}$ is $m \times 1$, $\mathbf{x}$ is $n \times 1$, $\mathbf{A}$ is $m \times n$, and $\mathbf{A}$ is independent of $\mathbf{x}$ and $\mathbf{y}$, then

$$\frac{\partial \alpha}{\partial \mathbf{x}} = \mathbf{y}^T \mathbf{A} \quad (36)$$

and

$$\frac{\partial \alpha}{\partial \mathbf{y}} = \mathbf{x}^\top \mathbf{A}^\top \quad (37)$$

Proof: Define

$$\mathbf{w}^\top = \mathbf{y}^\top \mathbf{A} \quad (38)$$

and note that

$$\alpha = \mathbf{w}^\top \mathbf{x} \quad (39)$$

Hence, by Proposition 5 we have that

$$\frac{\partial \alpha}{\partial \mathbf{x}} = \mathbf{w}^\top = \mathbf{y}^\top \mathbf{A} \quad (40)$$

which is the first result. Since α is a scalar, we can write

$$\alpha = \alpha^\top = \mathbf{x}^\top \mathbf{A}^\top \mathbf{y} \quad (41)$$

and applying Proposition 5 as before we obtain

$$\frac{\partial \alpha}{\partial \mathbf{y}} = \mathbf{x}^\top \mathbf{A}^\top \quad (42)$$

q.e.d.

Proposition 8 For the special case in which the scalar α is given by the quadratic form

$$\alpha = \mathbf{x}^\top \mathbf{A} \mathbf{x} \quad (43)$$

where $\mathbf{x}$ is $n \times 1$, $\mathbf{A}$ is $n \times n$, and $\mathbf{A}$ does not depend on $\mathbf{x}$, then

$$\frac{\partial \alpha}{\partial \mathbf{x}} = \mathbf{x}^\top (\mathbf{A} + \mathbf{A}^\top) \quad (44)$$

Proof: By definition

$$\alpha = \sum_{j=1}^n \sum_{i=1}^n a_{ij} x_i x_j \quad (45)$$

Differentiating with respect to the k th element of $\mathbf{x}$ we have

$$\frac{\partial \alpha}{\partial x_k} = \sum_{j=1}^n a_{kj} x_j + \sum_{i=1}^n a_{ik} x_i \quad (46)$$

for all $k = 1, 2, \dots, n$, and consequently,

$$\frac{\partial \alpha}{\partial \mathbf{x}} = \mathbf{x}^\top \mathbf{A}^\top + \mathbf{x}^\top \mathbf{A} = \mathbf{x}^\top (\mathbf{A}^\top + \mathbf{A}) \quad (47)$$

q.e.d.

Proposition 9 For the special case where $\mathbf{A}$ is a symmetric matrix and

$$\alpha = \mathbf{x}^T \mathbf{A} \mathbf{x} \quad (48)$$

where $\mathbf{x}$ is $n \times 1$, $\mathbf{A}$ is $n \times n$, and $\mathbf{A}$ does not depend on $\mathbf{x}$, then

$$\frac{\partial \alpha}{\partial \mathbf{x}} = 2\mathbf{x}^T \mathbf{A} \quad (49)$$

Proof: This is an obvious application of Proposition 8. q.e.d.

Proposition 10 Let the scalar α be defined by

$$\alpha = \mathbf{y}^T \mathbf{x} \quad (50)$$

where $\mathbf{y}$ is $n \times 1$, $\mathbf{x}$ is $n \times 1$, and both $\mathbf{y}$ and $\mathbf{x}$ are functions of the vector $\mathbf{z}$. Then

$$\frac{\partial \alpha}{\partial \mathbf{z}} = \mathbf{x}^T \frac{\partial \mathbf{y}}{\partial \mathbf{z}} + \mathbf{y}^T \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (51)$$

Proof: We have

$$\alpha = \sum_{j=1}^n x_j y_j \quad (52)$$

Differentiating with respect to the k th element of $\mathbf{z}$ we have

$$\frac{\partial \alpha}{\partial z_k} = \sum_{j=1}^n \left(x_j \frac{\partial y_j}{\partial z_k} + y_j \frac{\partial x_j}{\partial z_k} \right) \quad (53)$$

for all $k = 1, 2, \dots, n$, and consequently,

$$\frac{\partial \alpha}{\partial \mathbf{z}} = \frac{\partial \alpha}{\partial \mathbf{y}} \frac{\partial \mathbf{y}}{\partial \mathbf{z}} + \frac{\partial \alpha}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial \mathbf{z}} = \mathbf{x}^T \frac{\partial \mathbf{y}}{\partial \mathbf{z}} + \mathbf{y}^T \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (54)$$

q.e.d.

Proposition 11 Let the scalar α be defined by

$$\alpha = \mathbf{x}^T \mathbf{x} \quad (55)$$

where $\mathbf{x}$ is $n \times 1$, and $\mathbf{x}$ is a function of the vector $\mathbf{z}$. Then

$$\frac{\partial \alpha}{\partial \mathbf{z}} = 2\mathbf{x}^T \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (56)$$

Proof: This is an obvious application of Proposition 10. q.e.d.

Proposition 12 Let the scalar α be defined by

$$\alpha = \mathbf{y}^T \mathbf{A} \mathbf{x} \quad (57)$$

where $\mathbf{y}$ is $m \times 1$, $\mathbf{x}$ is $n \times 1$, $\mathbf{A}$ is $m \times n$, and both $\mathbf{y}$ and $\mathbf{x}$ are functions of the vector $\mathbf{z}$, while $\mathbf{A}$ does not depend on $\mathbf{z}$. Then

$$\frac{\partial \alpha}{\partial \mathbf{z}} = \mathbf{x}^T \mathbf{A}^T \frac{\partial \mathbf{y}}{\partial \mathbf{z}} + \mathbf{y}^T \mathbf{A} \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (58)$$

Proof: *Define*

$$\mathbf{w}^\top = \mathbf{y}^\top \mathbf{A} \quad (59)$$

and note that

$$\alpha = \mathbf{w}^\top \mathbf{x} \quad (60)$$

Applying Proposition 10 we have

$$\frac{\partial \alpha}{\partial \mathbf{z}} = \mathbf{x}^\top \frac{\partial \mathbf{w}}{\partial \mathbf{z}} + \mathbf{w}^\top \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (61)$$

Substituting back in for $\mathbf{w}$ we arrive at

$$\frac{\partial \alpha}{\partial \mathbf{z}} = \frac{\partial \alpha}{\partial \mathbf{y}} \frac{\partial \mathbf{y}}{\partial \mathbf{z}} + \frac{\partial \alpha}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial \mathbf{z}} = \mathbf{x}^\top \mathbf{A}^\top \frac{\partial \mathbf{y}}{\partial \mathbf{z}} + \mathbf{y}^\top \mathbf{A} \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (62)$$

q.e.d.

Proposition 13 *Let the scalar α be defined by the quadratic form*

$$\alpha = \mathbf{x}^\top \mathbf{A} \mathbf{x} \quad (63)$$

where $\mathbf{x}$ is $n \times 1$, $\mathbf{A}$ is $n \times n$, and $\mathbf{x}$ is a function of the vector $\mathbf{z}$, while $\mathbf{A}$ does not depend on $\mathbf{z}$. Then

$$\frac{\partial \alpha}{\partial \mathbf{z}} = \mathbf{x}^\top (\mathbf{A} + \mathbf{A}^\top) \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (64)$$

Proof: *This is an obvious application of Proposition 12.* q.e.d.

Proposition 14 *For the special case where $\mathbf{A}$ is a symmetric matrix and*

$$\alpha = \mathbf{x}^\top \mathbf{A} \mathbf{x} \quad (65)$$

where $\mathbf{x}$ is $n \times 1$, $\mathbf{A}$ is $n \times n$, and $\mathbf{x}$ is a function of the vector $\mathbf{z}$, while $\mathbf{A}$ does not depend on $\mathbf{z}$. Then

$$\frac{\partial \alpha}{\partial \mathbf{z}} = 2\mathbf{x}^\top \mathbf{A} \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \quad (66)$$

Proof: *This is an obvious application of Proposition 13.* q.e.d.

Definition 5 Let $\mathbf{A}$ be a $m \times n$ matrix whose elements are functions of the scalar parameter α . Then the derivative of the matrix $\mathbf{A}$ with respect to the scalar parameter α is the $m \times n$ matrix of element-by-element derivatives:

$$\frac{\partial \mathbf{A}}{\partial \alpha} = \begin{bmatrix} \frac{\partial a_{11}}{\partial \alpha} & \frac{\partial a_{12}}{\partial \alpha} & \dots & \frac{\partial a_{1n}}{\partial \alpha} \\ \frac{\partial a_{21}}{\partial \alpha} & \frac{\partial a_{22}}{\partial \alpha} & \dots & \frac{\partial a_{2n}}{\partial \alpha} \\ \vdots & \vdots & & \vdots \\ \frac{\partial a_{m1}}{\partial \alpha} & \frac{\partial a_{m2}}{\partial \alpha} & \dots & \frac{\partial a_{mn}}{\partial \alpha} \end{bmatrix} \quad (67)$$

Proposition 15 *Let $\mathbf{A}$ be a nonsingular, $m \times m$ matrix whose elements are functions of the scalar parameter α . Then*

$$\frac{\partial \mathbf{A}^{-1}}{\partial \alpha} = -\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial \alpha} \mathbf{A}^{-1} \quad (68)$$

Proof: *Start with the definition of the inverse*

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (69)$$

and differentiate, yielding

$$\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial \alpha} + \frac{\partial \mathbf{A}^{-1}}{\partial \alpha}\mathbf{A} = \mathbf{0} \quad (70)$$

rearranging the terms yields

$$\frac{\partial \mathbf{A}^{-1}}{\partial \alpha} = -\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial \alpha}\mathbf{A}^{-1} \quad (71)$$

q.e.d.

6 References

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Regression Analysis Linear Algebra

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Simple linear regression with Vector notation

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- Consider a data set $D = \{(x_i, y_i) | x_i \in \mathbb{R}, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$
- x_i s are non stochastic
- y_i s are stochastic and realized values of random variable Y_i s
- $\mathbf{y} = (y_1, y_2, \dots, y_n)^T$, $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$, $\boldsymbol{\beta} = (\beta_0, \beta_1)^T$ and $\mathbf{1} = (1, 1, \dots, 1)^T$

Problem statement (Redefined)

We are interested to have a prediction vector

$$\hat{\mathbf{y}} = g(\mathbf{x}, \boldsymbol{\beta}) = [\mathbf{1} \ \mathbf{x}] \boldsymbol{\beta}$$

which will approximate well the observed vector $\mathbf{y}$ for known vector $\mathbf{x}$.

It is a problem in $\mathbb{R}^n$ now !!



Other uses of vector representation

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- Weighted sum / Averaging
- Expectation of discrete random variable
- Combing audio signals for music composition
- Image representation in pic-cell.
- Principal component Analysis
- $\mathbb{P}_n$ = Polynomial up to degree n



Vector Space $(V, +, \cdot)$

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Definition

A vector space V over real numbers $\mathbb{R}$ is a collection of vectors such that

- 1** $+$: $V \times V \rightarrow V$ [closed under vector addition]
- 2** $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$, for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ [associative]
- 3** There exists $\mathbf{0} \in V$ such that
 $\mathbf{0} + \mathbf{x} = \mathbf{x} + \mathbf{0} = \mathbf{x}$ for all $\mathbf{x} \in V$ [identity element exists]
- 4** There exists $-\mathbf{x} \in V$ for each $\mathbf{x}$ such that
 $(-\mathbf{x}) + \mathbf{x} = \mathbf{x} + (-\mathbf{x}) = \mathbf{0}$ [inverse exists]
- 5** $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$ [commutative]
- 6** $a \cdot (b \cdot \mathbf{x}) = (ab) \cdot \mathbf{x}$ for all $a, b \in \mathbb{R}$ and $\mathbf{x} \in V$
- 7** $1 \cdot \mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in V$
- 8** $(a + b) \cdot \mathbf{x} = (a \cdot \mathbf{x}) + (b \cdot \mathbf{x})$ for all $a, b \in \mathbb{R}$ and $\mathbf{x} \in V$
- 9** $a \cdot (\mathbf{x} + \mathbf{y}) = a \cdot \mathbf{x} + a \cdot \mathbf{y}$



Sub-Space $(S, +, \cdot)$

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Definition

If a subset S of V is a vector space itself then S is called subspace of V .

How to check S is a subspace of V ?

- (1) Whether $\mathbf{0} \in S$?
- (2) Whether $\mathbf{x} + a \cdot \mathbf{y} \in S$? for all $\mathbf{x}, \mathbf{y} \in S$ and $a \in \mathbb{R}$.

Example:

- (1) All lines passing through $(0, 0)$ in $\mathbb{R}^2$.
- (2) All planes passing through origin in $\mathbb{R}^n$.
- (3) $\mathbb{P}_5$ in $\mathbb{P}_7$



Definition

The span of a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} \in \mathbf{V}$ is the collection

$$Sp\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} = \left\{ \sum_{i=1}^k c_i \mathbf{v}_i \mid c_i \in \mathbb{R} \right\}$$

which is the collection of all possible linear combinations of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$.

Note: A span is always a subspace.

Example :

(a) $Sp\{(0, 1), (1, 1)\} = Sp\{(0, 1), (1, 0)\} = \mathbb{R}^2$

(b) $Sp\{(0, 1, 0), (1, 1, 0)\} = \mathbb{R} \times \mathbb{R} \times \{0\} = xy\text{-pane in } \mathbb{R}^3$

In regression $\hat{\mathbf{y}} \in Sp\{\mathbf{1}, \mathbf{x}\}$ which is closest to $\mathbf{y} \in \mathbb{R}^n$



Linear Independence

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Definition

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} \in \mathbf{V}$ are said to be linearly independent iff $\sum_{i=1}^k c_i \mathbf{v}_i = \mathbf{0} \implies c_1 = c_2 = \dots = c_n = 0$. On the other hand if $\sum_{i=1}^k c_i \mathbf{v}_i = \mathbf{0}$ holds for some non zero $c_i \in \mathbb{R}$ the the vectors are called linearly dependent.

Example :

- (a) $\{(0, 1), (1, 1)\}$ are independent
- (b) $\{(0, 1), (1, 0)\}$ are independent
- (c) $\{(0, 1), (1, 0), (1, 1)\}$ are dependent



Basis

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Definition

If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ are linearly independent then it is a basis of $Sp\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$, and the dimension of $Sp\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is the number of linearly independent elements in $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$.

Example :

- (a) $\{(0, 1), (1, 1)\}$ is a basis of $\mathbb{R}^2$
- (b) $\{(0, 1), (1, 0)\}$ is a basis of $\mathbb{R}^2$ also.
- (c) $\{(0, 1), (1, 0), (1, 1)\}$ is NOT a basis of $\mathbb{R}^2$

Note: Number of vectors in a basis of a vector space is known as the dimension of the vector space.



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Orthogonal vectors

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbf{V}$ are said to be orthogonal if $\mathbf{u}^T \mathbf{v} = \sum_i u_i v_i = 0$

Orthogonal complement

If $\mathbf{S} \subseteq \mathbf{V}$ is a subspace then the orthogonal complement of $\mathbf{S}$ denoted by $\mathbf{S}^\perp$ is a collection

$$\mathbf{S}^\perp = \{\mathbf{v} | \mathbf{v} \in \mathbf{V}, \mathbf{u}^T \mathbf{v} = 0, \forall \mathbf{u} \in \mathbf{S}\}$$

and $\dim(\mathbf{S}^\perp) = \dim(\mathbf{V}) - \dim(\mathbf{S})$.

Examples:

(a) $\text{Sp}\{(1, 0, 0, 0), (0, 0, 1, 0)\} \perp \text{Sp}\{(0, 1, 0, 0), (0, 0, 0, 1)\}$

(b) $\text{Sp}\{(1, 1, 0, 0), (0, 1, 1, 0), (1, 0, 1, 0)\} \perp \text{Sp}\{(0, 0, 0, 1)\}$



Remarks

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- 1 Basis is not unique.
- 2 Elements of a basis are need not be orthogonal to each other.
- 3 Linear independence need not imply orthogonality.
- 4 Orthogonality implies independence.
- 5 Orthogonal vectors with unit length are called orthonormal vectors.



Projection

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Projection Matrix

If $\mathbf{S} \subseteq \mathbf{V}$ then the projection matrix of subspace $\mathbf{S}$ is P_s satisfying

- (a) $P_s \mathbf{v} = \mathbf{v}$ if $\mathbf{v} \in \mathbf{S}$
- (b) $P_s \mathbf{v} \in \mathbf{S}$ for all $\mathbf{v} \in \mathbf{V}$

Orthogonal Projection Matrix

A projection matrix P_s is an orthogonal projection matrix of subspace $\mathbf{S} \subseteq \mathbf{V}$ if $(\mathbf{I} - P_s)$ is a projection matrix of $\mathbf{S}^\perp \subseteq \mathbf{V}$ too.

Theorem

If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is an orthonormal basis of the subspace $\mathbf{S} \subseteq \mathbf{V}$ then the orthogonal projection matrix of $\mathbf{S}$ is $P_s = \sum_{i=1}^k \mathbf{v}_i \mathbf{v}_i^T$



Idempotency

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Idempotent matrix

If a matrix P satisfies the relation that $P^2 = P$, then P is called an idempotent matrix.

Theorem

An idempotent matrix has eigen values 0 and 1.

Theorem

A projection matrix is an idempotent matrix.

In regression eventually $\hat{\mathbf{y}}$ becomes
the orthogonal projection of $\mathbf{y} \in \mathbb{R}^n$ in the subspace $S = Sp \{ \mathbf{1}, \mathbf{x} \}$



Column Space

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Definition

The column space of a matrix $A = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n]$ with columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ is

$$\mathcal{C}(A) = \text{Sp}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\}.$$

Hence, row-space of A denoted by $\mathcal{R}(A) = \mathcal{C}(A^T)$.

Properties :

- 1 $\mathcal{C}(A : B) = \mathcal{C}(A) + \mathcal{C}(B)$
- 2 $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$
- 3 $\dim(\mathcal{C}(A)) = \text{Rank}(A)$
- 4 $\mathcal{C}(AA^T) = \mathcal{C}(A) \implies \text{Rank}(AA^T) = \text{Rank}(A)$
- 5 If A has n -rows then $\dim(\mathcal{C}(A)^\perp) = n - \text{Rank}(A)$



Quadratic forms

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Definition

A square symmetric matrix $\mathbf{A} = ((A_{ij}))_{n \times n}$ is said to be

(a) **positive definite (p.d.)** if

$$\mathbf{x}^T \mathbf{A} \mathbf{x} > 0 \text{ for all } \mathbf{x} \neq \mathbf{0} \in \mathbb{R}^n.$$

(b) **positive semi-definite (p.s.d)** if

$$\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0 \text{ for all } \mathbf{x} \neq \mathbf{0} \in \mathbb{R}^n.$$

[Also called non-negative definite (n.n.d.)]

Properties:

(a) If $\mathbf{A}$ is p.d. then $|\mathbf{A}| > 0$.

(b) If $\mathbf{A}$ is p.s.d. then $|\mathbf{A}| \geq 0$.



Generalized inverse

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Definition

A matrix G is said to be a generalize inverse of a matrix A if $AGA = A$. Usually G is denoted by A^- .

Properties:

- (1) If A is $m \times n$ then A^- is $n \times m$.
- (2) A^- is not unique.
- (3) For a matrix A the projection matrix of $\mathcal{C}(A)$ is AA^-
- (4) For a matrix A **the orthogonal projection matrix of $\mathcal{C}(A)$ is**

$$A(A^T A)^- A^T.$$

In regression the prediction $\hat{\mathbf{y}}$ and the error $\mathbf{y} - \hat{\mathbf{y}}$ are orthogonal to each other.



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10. ESTIMATION

Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ be the observed/ realized values of a set of i.i.d. random variables $\mathbf{X} = (X_1, X_2, \dots, X_n)$ where $X_i \stackrel{iid}{\sim} f_\theta$ for some $\theta \in \Theta$. Here a family of distributions is denoted by

$$\mathcal{F} = \{f(x|\theta)|\theta \in \Theta\} \quad \text{or} \quad \{F(x|\theta)|\theta \in \Theta\}$$

Parametric Estimation: In a parametric inference problem it is assumed that the family of the distribution is known but the particular value of the parameter is unknown. We estimate the value of the parameter θ as a function of the observations $\mathbf{x}$. The ultimate goal is to approximate the p.d.f f_θ or F_θ through the estimation of θ itself. Parametric estimation has two aspects, namely, (a) **Point estimation** and (b) **Interval estimation**. [We will learn it after Testing]

In point estimation we will learn

- (a) Definition of an estimator
- (b) Good properties of an estimator
- (c) Methods of estimation (MME and MLE)

Definition 129. Statistic: A statistic is a function of random variables and it is free from any unknown parameter. Being a (measurable) function, $T(\mathbf{X})$ say, of random variables it is also a random variable.

Definition 130. Estimator: If the statistic $T(\mathbf{X})$ is used to estimate a parametric function $g(\theta)$ then $T(\mathbf{X})$ is said to be {an estimator of $g(\theta)$ }. And a realized value of it for $\mathbf{X} = \mathbf{x}$ i.e. $T(\mathbf{x})$ is known as **an estimate** of θ . We often abuse the notation as $g(\hat{\theta}) = T(\mathbf{x})$ and $g(\hat{\theta}) = T(\mathbf{X})$ which are understood from the context.

Definition 131. Unbiased estimator: An estimator $T(\mathbf{X})$ is said to be an unbiased estimator of a parametric function $g(\theta)$ if $E(T(\mathbf{X}) - g(\theta)) = 0 \forall \theta \in \Theta$.

Remark 132. It does not require $T(\mathbf{x}) = g(\theta)$ to hold or it may hold with probability zero.

Definition 133. Bias: The bias of an estimator $T(\mathbf{X})$ while estimating a parametric function $g(\theta)$ is $B_{g(\theta)}(T(\mathbf{X})) = E(T(\mathbf{X}) - g(\theta)) \forall \theta \in \Theta$.

Definition 134. Asymptotically unbiased estimator: Denoting $T_n = T(X_1, X_2, \dots, X_n)$ an estimator T_n is said to be asymptotically unbiased of $g(\theta)$ if

$$\lim_{n \rightarrow \infty} B_{g(\theta)}(T_n) = \lim_{n \rightarrow \infty} E(T_n - g(\theta)) = 0$$

Definition 135. Consistent estimator: An estimator T_n is said to be consistent estimator $g(\theta)$ if $T_n \xrightarrow{P} g(\theta)$ i.e.

$$\lim_{n \rightarrow \infty} P(|T_n - g(\theta)| < \epsilon) = 1 \forall \theta \in \Theta, \epsilon > 0$$

Definition 136. Mean squared error (MSE): The MSE of an estimator $T(\mathbf{X})$ while estimating a parametric function $g(\theta)$ is

$$MSE_{g(\theta)}(T(\mathbf{X})) = E[(T(\mathbf{X}) - g(\theta))^2] \forall \theta \in \Theta.$$

Exercise 137. Show that $MSE_{g(\theta)}(T(\mathbf{X})) = Var(T(\mathbf{X})) + B_{g(\theta)}^2(T(\mathbf{X}))$

Exercise 138. If $MSE_{g(\theta)}(T_n(\mathbf{X})) \downarrow 0$ as $n \uparrow \infty$ then show that $(T_n(\mathbf{X}))$ is a consistent estimator.

Remark 139. Asymptotic unbiasedness and consistency are large sample properties and both are based on L_1 norm. . MSE is defined based on L_2 norm.

Exercise 140. Let $(X_1, X_2, \dots, X_n)$ be i.i.d random variables with $E(X) = \mu$ and $Var(X) = \sigma^2$. and define $T_n(\mathbf{X}) = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, $S_1^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ and $S_2^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. Show that

- (a) $T_n(\mathbf{X})$ is an unbiased estimator of μ .
- (b) S_1^2 is a biased estimator of σ^2
- (c) S_2^2 is an asymptotically unbiased estimator of σ^2

Exercise 141. Let $(X_1, X_2, \dots, X_n) \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Show that $MSE(S_2^2) < MSE(S_1^2)$.
Note: Unbiased estimator need not have minimum MSE.

Definition 142. Method of Moment for Estimation (MME): Consider $\mathbf{x} = (x_1, x_2, \dots, x_n)$ be the observed/ realized values of a set of i.i.d. random variables

$\mathbf{X} = (X_1, X_2, \dots, X_n)$ where $X_i \stackrel{iid}{\sim} f_\theta$ for some $\theta \in \Theta$. Then

Step 1: Computer theoretical moments from the p.d.f.

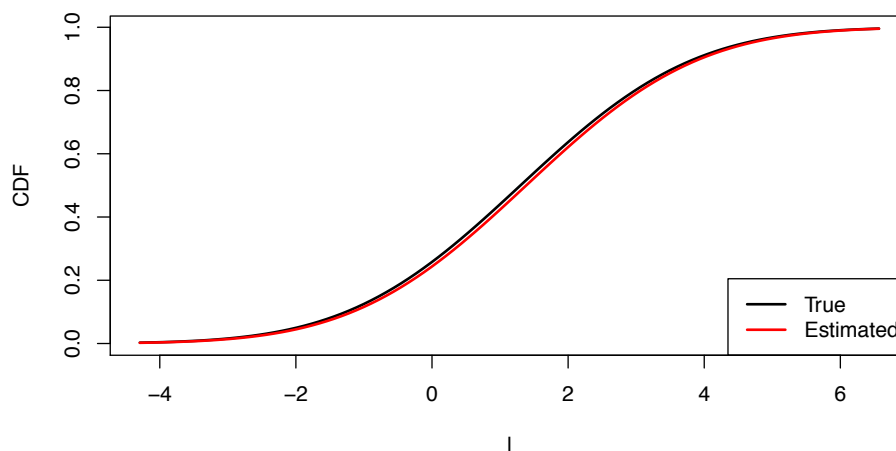
Step 2: Computer empirical moments from the data.

Step 3: Construct k equations if you have k unknown parameters.

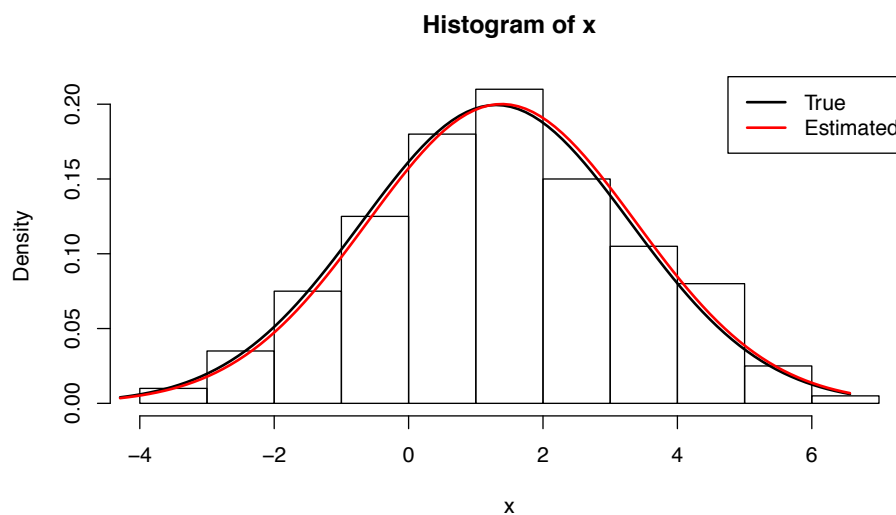
Step 4: Solve the equations for the parameters.

```
# Distribution : Normal
mu<-1.3 # mean
s<- 2 # sigma
n<- 200 # sample size
x<- rnorm(n,mean = mu,sd = s) # data
xmin<- min(x) # min of data
xmax<-max(x) # max data
l<- seq(xmin-0.5, xmax+0.5, length=100)
##### Estimation #####
muh<-mean(x)
sh<-sd(x)
#####
cat("True mean=", mu, "estimated mean=", muh, "\n")
## True mean= 1.3 estimated mean= 1.385195
cat("True sigma=", s, "estimated sigma=", sh, "\n")
## True sigma= 2 estimated sigma= 1.993788
#####

plot(pnorm(q = l,mean = mu,sd = s)~l, type = 'l', col=1, lwd=2, ylab = "CDF")
lines(pnorm(q = l,mean = muh,sd = sh)~l, type = 'l', col=2, lwd=2)
legend("bottomright",legend = c("True", "Estimated"), col = c(1,2), lwd = c(2,2))
```



```
hist(x,probability = T)
lines(dnorm(x = l,mean = mu,sd = s)~l, type = 'l', col=1, lwd=2, ylab = "PDF")
lines(dnorm(x=l,mean = muh,sd = sh)~l, type = 'l', col=2, lwd=2)
legend("topright",legend = c("True", "Estimated"), col = c(1,2), lwd = c(2,2))
```



Remark 143. We can not use MME to estimate the parameters of $C(\mu, \sigma)$, because the moments does not exists for Cauchy distribution.

Definition 144. Maximum Likelihood Estimator: Consider $\mathbf{x} = (x_1, x_2, \dots, x_n)$ be the observed/ realized values of a set of i.i.d. random variables $\mathbf{X} = (X_1, X_2, \dots, X_n)$ where $X_i \stackrel{iid}{\sim} f_\theta$ for some $\theta \in \Theta$. Then the joint p.d.f. of $\mathbf{X} = (X_1, X_2, \dots, X_n)$ is a

function of $\mathbf{x}$ when the parameter value is fixed i.e.

$$f(\mathbf{x}|\theta) = \prod_{i=1}^n f(x_i, \theta)$$

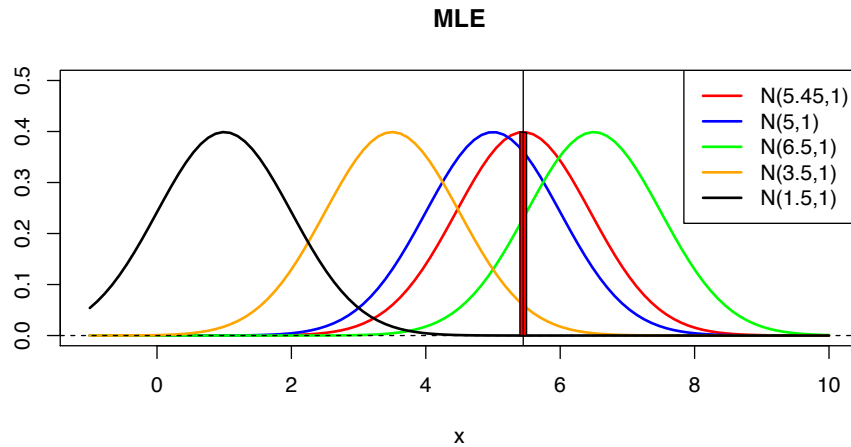
and the likelihood of a function of parameter for a given set of data $\mathbf{X} = \mathbf{x}$ i.e.

$$\ell(\theta|\mathbf{x}) = \prod_{i=1}^n f(x_i, \theta).$$

Hence the maximum likelihood estimator of θ is

$$\hat{\theta}_{mle} = \arg \max_{\theta \in \Theta} \ell(\theta|\mathbf{x}) = \arg \max_{\theta \in \Theta} \log \ell(\theta|\mathbf{x})$$

NOTE: Finding the maxima through differentiation is possible **only** if ℓ is a smoothly differentiable function w.r.t θ . Otherwise it has to be maximized by some other methods. **Differentiation is not the only way of finding maxima or minima.**



Exercise 145. $(X_1, X_2, \dots, X_n) \stackrel{iid}{\sim} U(0, \theta)$. where $\theta \in \Theta = (0, \infty)$.

- Find the MLE of θ .
- Is it an unbiased estimator ?
- Find the MSE.

Exercise 146. Let $(X_1, X_2, \dots, X_n) \stackrel{iid}{\sim} N(\mu, \sigma^2)$.

- Find the *MME* and *MLE* of μ and σ^2 . Are they same ?
- Are they unbiased estimators?

Exercise 147. Let $(X_1, X_2, \dots, X_n) \stackrel{iid}{\sim} \text{Gamma}(\alpha, \lambda)$.

- Find the *MME* of (α, λ) ?
- Find MLE of (α, λ) by an iterative method of solution.

NOTE: You may use the MME as an initial value of iteration to obtain the MLE.

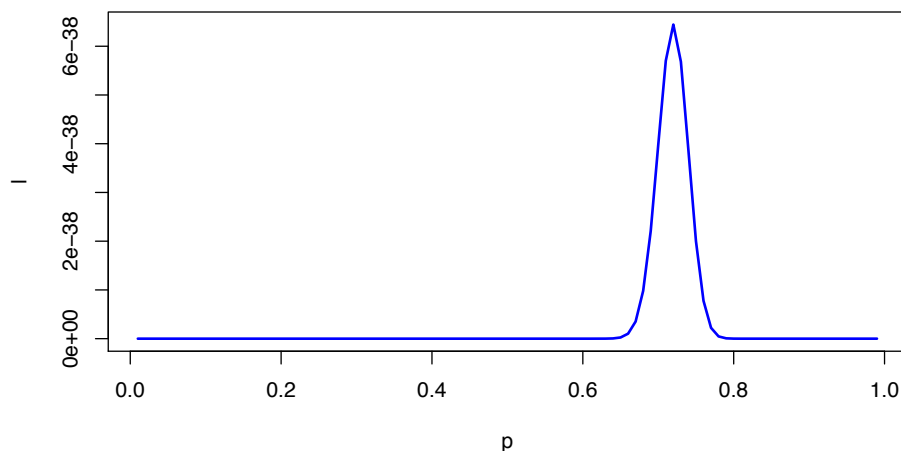
```

# MLE of binomial parameter
set.seed(12)
n<-10 # size of binomial
x<- sort(rbinom (50, n, 0.7)) # sample given
print(x)
## [1] 4 5 5 5 5 6 6 6 6 6 6 6 6 6 6 7 7 7 7 7 7 7
## [24] 7 7 7 7 7 8 8 8 8 8 8 8 8 8 8 8 8 8 9 9 9 9
## [47] 9 9 10 10

# MLE finding
p<-seq(0.01,0.99,by = 0.01)
l<-array(0,dim=c(length(p)))
for (i in 1 : length(p)){
  l[i]<-prod(dbinom(x,n,p[i])) # product of likelihood
}

plot(l~p, type='l', col=4, lwd=2)

```



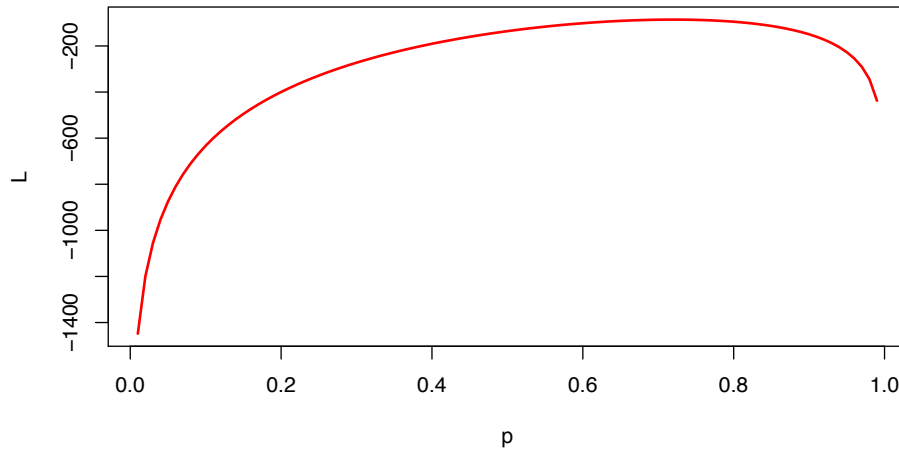
```

mle1<-p[which(l==max(l))]
print(mle1)
## [1] 0.72

L<-array(0,dim=c(length(p)))
for (i in 1 : length(p)){
  L[i]<-sum(log(dbinom(x,n,p[i]))) #sum of log likelihood
}

plot(L~p,type='l', col=2, lwd=2)

```



```
mle2<-p[which(L==max(L))]  
print(mle2)  
## [1] 0.72
```

Properties of MLE:

- (a) MLE need not be unique.
- (b) MLE need not be an unbiased estimator.
- (c) MLE is always a consistent estimator.
- (d) MLE is asymptotically normally distributed up to some location and scale when some regularity condition satisfied like
 - (1) Range of the random variable is free from parameter.
 - (2) Likelihood is smoothly differentiable for up to 3rd order and corresponding expectations exists.

Exercise 148. $(X_1, X_2, \dots, X_n) \stackrel{iid}{\sim} U(\theta - 0.5, \theta + 0.5)$ where $\theta \in \Theta = (-\infty, \infty)$.

- (a) Find the MLE of θ .
- (b) Is it unique?
- (c) Is it consistent? Find the MSE.

Definition 149. Interval Estimation: Consider a pair of statistic $(L(\mathbf{X}), U(\mathbf{X}))$ such that for a parameter θ ,

$$P_{\theta}(\theta \in [L(\mathbf{X}), U(\mathbf{X})]) = 1 - \alpha$$

Then a $100(1 - \alpha)\%$ confidence interval of θ is considered to be $[L(\mathbf{X}), U(\mathbf{X})]$.

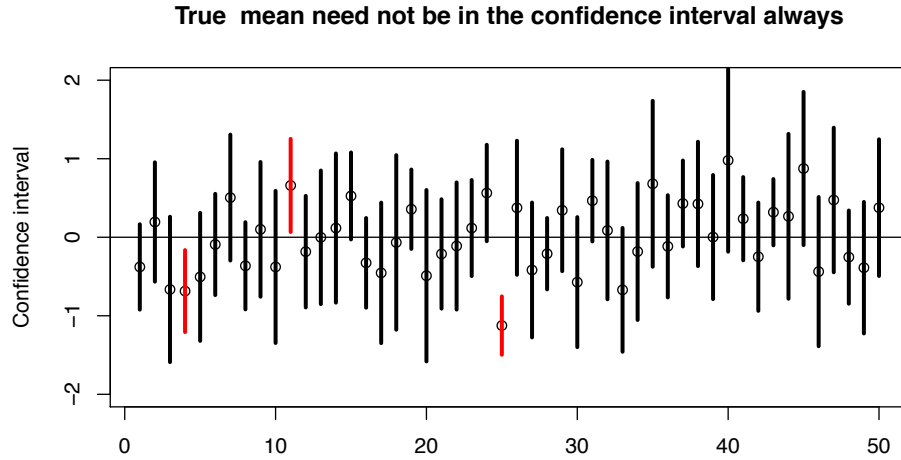
Example 150. If $X_1, X_2, \dots, X_n$ are i.i.d random variables with $N(\mu, \sigma^2)$ distribution with known value of σ^2 . Then a $100(1 - \alpha)\%$ CI of μ is

$$\left[L(\mathbf{X}) = \bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, U(\mathbf{X}) = \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \right]$$

```

set.seed(10)
N <- 50
n <- 8 # sample size
v <- matrix(c(0,0),nrow=2)
for (i in 1:N) {
  x <- rnorm(n)
  v <- cbind(v, t.test(x)$conf.int)
}
v <- v[,2:(N+1)]
plot(apply(v,2,mean), ylim=c(-2,2), ylab='Confidence interval', xlab='')
abline(0,0)
c <- apply(v,2,min)>0 | apply(v,2,max)<0
segments(1:N,v[1,],1:N,v[2,], col=c(par('fg'),'red')[c+1], lwd=3)
title(main="True mean need not be in the confidence interval always")

```



Example 151. If $X_1, X_2, \dots, X_n$ are i.i.d random variables with $N(\mu, \sigma^2)$ distribution. Then a $100(1 - \alpha)\%$ CI of μ is

$$\left[L(\mathbf{X}) = \bar{X} - \frac{\hat{\sigma}_u}{\sqrt{n}} \tau_{\alpha/2, n-1}, U(\mathbf{X}) = \bar{X} + \frac{\hat{\sigma}_u}{\sqrt{n}} \tau_{\alpha/2, n-1} \right]$$

$\hat{\sigma}_u^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of unknown variance and a $100(1 - \alpha)\%$ CI of σ^2 is

$$\left[L(\mathbf{X}) = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\chi_{\alpha/2, (n-1)}^2}, U(\mathbf{X}) = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\chi_{1-\alpha/2, (n-1)}^2} \right]$$

11. TESTING OF HYPOTHESIS

Definition 152. Hypothesis: A hypothesis in parametric inference is a statement about the population parameter. It has two categories. A **null hypothesis** (H_0) specifies a subset Θ_0 in the parameter space Θ . If Θ_a is a singleton set then it called a **simple null**, otherwise a **composite null**. On the other hand an **alternative hypothesis** (H_1) specifies another subset $\Theta_a \subset \Theta$ which is disjoint to Θ_0 .

Definition 153. Test Rule: A test rule is a statistical procedure, based on the distribution of the test statistic, which will reject the null hypothesis in favour of the alternative hypothesis.

Definition 154. Rejection Region or Critical region: A rejection Region or critical region is a subset $C \subset \mathbb{R}^n$ such that $\mathbf{X} \in C \Leftrightarrow T(\mathbf{X})$ will reject the null hypothesis.

Definition 155. Level- α test: For any $\alpha \in (0, 1)$, a test is said to be level- α test if

$$\sup_{\theta \in \Theta_0} P_{\theta}(\mathbf{X} \in C) \leq \alpha.$$

Definition 156. Size- α test: For any $\alpha \in (0, 1)$, a test is said to be size- α test if

$$\sup_{\theta \in \Theta_0} P_{\theta}(\mathbf{X} \in C) = \alpha.$$

Definition 157. Power-function: A power function is a function

$$P_{\theta}(\mathbf{X} \in C) : \Theta_a \rightarrow [0, 1]$$

Remark 158. More than one tests with same level can be compared in terms of power functions. A test procedure with more power than the other with same level can be considered a better test.

Definition 159. Type-I error: The event $\mathbf{X} \in C$ when $\theta \in \Theta_0$ is known as Type-I error.

Definition 160. Type-II error: The event $\mathbf{X} \in C^c$ when $\theta \in \Theta_a$ is known as Type-II error. Power is 1-P(Type-II error).

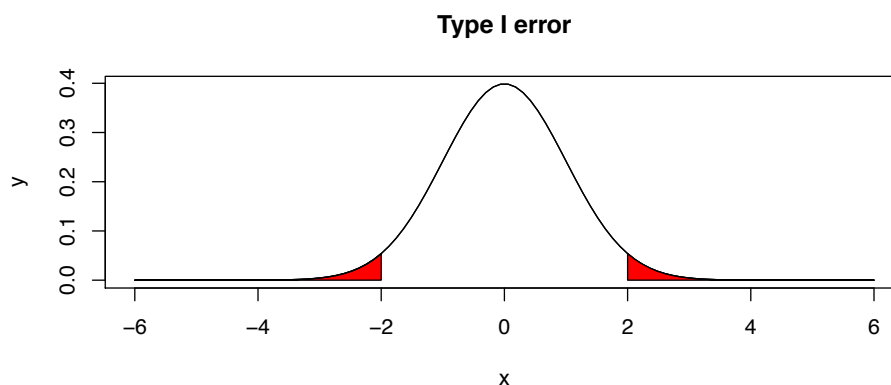
```
colorie <- function (x, y1, y2, N=1000, ...) {
  for (t in (0:N)/N) {
    lines(x, t*y1+(1-t)*y2, ...)
  }
}
# No, there is already a function to do this
colorie <- function (x, y1, y2, ...) {
  polygon( c(x, x[length(x):1]), c(y1, y2[length(y2):1]), ... )
}
x <- seq(-6,6, length=100)
y <- dnorm(x)
plot(y~x, type='l')
i = x<qnorm(.025)
colorie(x[i],y[i],rep(0,sum(i)) ,col='red')
i = x>qnorm(.975)
```



```

colorie(x[i],y[i],rep(0,sum(i)) ,col='red')
lines(y~x)
title(main="Type I error")

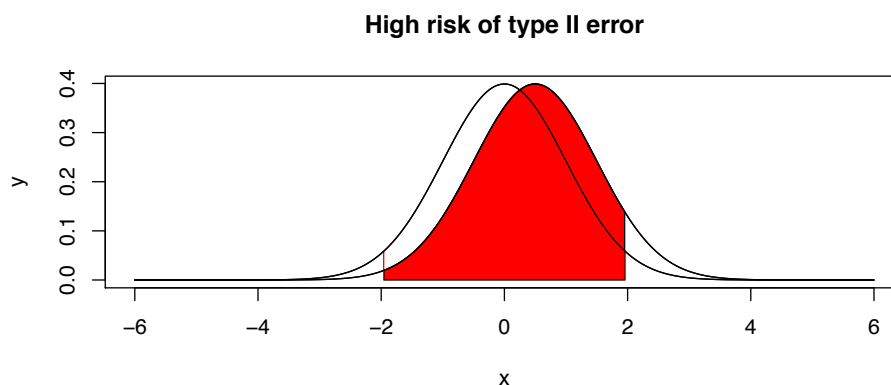
```



```

x <- seq(-6,6, length=1000)
y <- dnorm(x)
plot(y~x, type='l')
y2 <- dnorm(x-.5)
lines(y2~x)
i <- x>qnorm(.025) & x<qnorm(.975)
colorie(x[i],y2[i],rep(0,sum(i)), col='red')
segments( qnorm(.025),0,qnorm(.025),dnorm(qnorm(.025)), col='red' )
segments( qnorm(.975),0,qnorm(.975),dnorm(qnorm(.975)), col='red' )
lines(y~x)
lines(y2~x)
title(main="High risk of type II error")

```



```

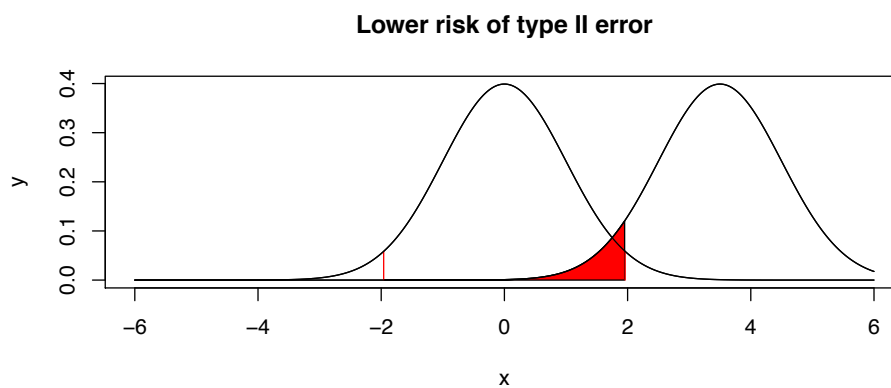
x <- seq(-6,6, length=1000)
y <- dnorm(x)

```

```

plot(y~x, type='l')
y2 <- dnorm(x-3.5)
lines(y2~x)
i <- x>qnorm(.025) & x<qnorm(.975)
colorie(x[i],y2[i],rep(0,sum(i)), col='red')
segments( qnorm(.025),0,qnorm(.025),dnorm(qnorm(.025)), col='red' )
segments( qnorm(.975),0,qnorm(.975),dnorm(qnorm(.975)), col='red' )
lines(y~x)
lines(y2~x)
title(main="Lower risk of type II error")

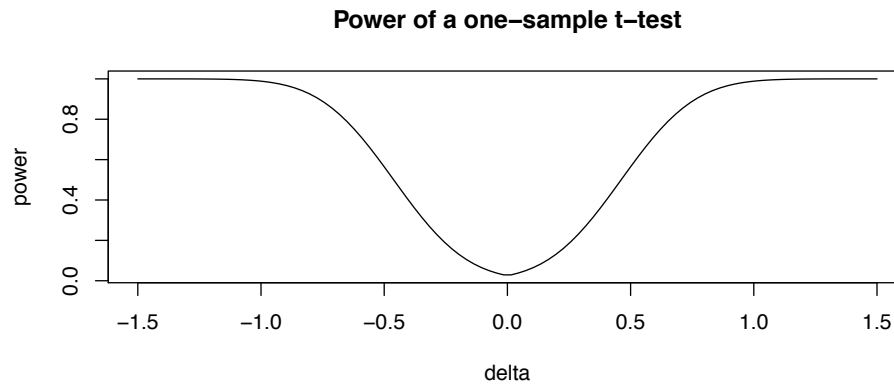
```



```

delta <- seq(-1.5, 1.5, length=100)
p <- NULL
for (d in delta) {
  p <- append(p,
              power.t.test(delta=d, sd=1, sig.level=0.05, n=20,
                           type='one.sample')$power)
}
plot(p~delta, type='l',
     ylab='power', main='Power of a one-sample t-test')

```



Lemma 161. Neyman-Pearson Lemma (1933): To test $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ reject H_0 in favour of H_1 at level/ size α if

$$\Lambda(\mathbf{x}) = \frac{f(\mathbf{x}|\theta_0)}{f(\mathbf{x}|\theta_1)} \leq \xi \quad \text{such that} \quad P_{\theta_0}(\Lambda(\mathbf{X}) \leq \xi) = \alpha$$

How to perform a test ??

Step1: Estimate the parameter for which the testing to be done.

Step2: Estimate the unknown parameters if any.

Step3: Construct the test statistic and obtain its value.

Step4: Obtain the exact or asymptotic distribution of the test statistic under the null hypothesis.

Step5: Depending on the alternative hypothesis (H_1) and level (α) decide the cut-off value or rejection condition.

Step6: Compare the observed value of test statistic (from Step 4) and the cut off value (from Step 5) to conclude the test. You may use **p-value** also.

Exercise 162. Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ Perform a test at size 0.05 for

(a) $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$. when σ^2 is known

(b) $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$. when σ^2 is unknown

(a) $H_0 : \sigma^2 = \sigma_0^2$ vs $H_1 : \sigma^2 \neq \sigma_0^2$ when μ is unknown

```
library("TeachingDemos")
n<-10
mu_true<-10.5
sd_true<-1.2
x<-rnorm(10,mu_true,sd_true) # generate data
#####
print(x)
## [1] 10.404652 11.918102 13.123373 10.987410 9.613969 8.152216 8.159945
## [8] 9.370803 11.937344 9.750913
cat("Unbiased estimate of mean =",mean(x), "\n")
## Unbiased estimate of mean = 10.34187
cat("Unbiased estimate of variance =",var(x), "\n")
```

```

## Unbiased estimate of variance = 2.729432
alpha<-0.05
## (a)H_0: mu = 10 vs H_1: mu not equal to 10 when sigma^2 = (1.2)^2 is known
za<-z.test(x,mu = 10,stdev = sd_true ,alternative =c("two.sided"),conf.level = (1-alpha))
print(za)
##
## One Sample z-test
##
## data: x
## z = 0.90091, n = 10.00000, Std. Dev. = 1.20000, Std. Dev. of the
## sample mean = 0.37947, p-value = 0.3676
## alternative hypothesis: true mean is not equal to 10
## 95 percent confidence interval:
## 9.598119 11.085627
## sample estimates:
## mean of x
## 10.34187
##(b)H_0: mu = 10 vs H_1: mu not equal to 10 when sigma^2 is unknown
ta<-t.test(x, mu = 10,alternative =c("two.sided"),conf.level = (1-alpha))
print(ta)
##
## One Sample t-test
##
## data: x
## t = 0.65438, df = 9, p-value = 0.5292
## alternative hypothesis: true mean is not equal to 10
## 95 percent confidence interval:
## 9.160032 11.523713
## sample estimates:
## mean of x
## 10.34187
##(c)H_0: sigma^2 = 1 vs H_0: sigma^2 neq 1 when mu is unknown
va<-sigma.test(x, sigma = 1,alternative = "two.sided", conf.level = (1-alpha))
print(va)
##
## One sample Chi-squared test for variance
##
## data: x
## X-squared = 24.565, df = 9, p-value = 0.006984
## alternative hypothesis: true variance is not equal to 1
## 95 percent confidence interval:
## 1.291341 9.096795
## sample estimates:
## var of x
## 2.729432

```

Exercise 163. Let $X_1, \dots, X_n \sim N(\mu_1, \sigma^2)$ (iid) and $Y_1, \dots, Y_m \sim N(\mu_2, \sigma^2)$ (iid) are independent. Perform a test at size 0.05 for $H_0 : \mu_1 = \mu_2$ vs $H_1 : \mu_1 \neq \mu_2$.

Exercise 164. Let $(X_i, Y_i) \sim N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$, $i = 1, 2, \dots, n$. Perform a test at size 0.05 for $H_0 : \mu_x = \mu_y$ vs $H_1 : \mu_x \neq \mu_y$. (this is known as paired-T test).

Exercise 165. Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$ Perform a test for $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$ at size 0.05.

Exercise 166. Let $X_1, \dots, X_n \sim N(\mu_1, \sigma_1^2)$ (iid) and $Y_1, \dots, Y_m \sim N(\mu_2, \sigma_2^2)$ (iid) are independent. Perform a test at size 0.05 for $H_0 : \sigma_1^2 = \sigma_2^2$ vs $H_1 : \sigma_1^2 \neq \sigma_2^2$.

List of Test Statistic: http://en.wikipedia.org/wiki/Test_statistic

- (1) Mathematical Statistics and Data Analysis by John A. Rice
- (2) Probability and Statistical Inference by Hogg, R. V., Tanis, E. A. & Zimmerman D. L.

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Multivariate
Analysis

B Banerjee

Expectation

Dispersion

Multivariate
Normal

Quadratic forms

T-Statistic

Regression Analysis Multivariate Analysis

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Expectation

Multivariate
Analysis

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Expectation

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T-Statistic

Definition

Let $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$ be a random vector with finite expectation for each of the component the we define expectation of a random vector as

$$E(\mathbf{X}) = (E(X_1), E(X_2), \dots, E(X_n))^T.$$

Similarly if $\mathbf{Y} = ((Y_{ij}))_{m \times n}$ is a random matrix with finite expectation for each of the component the we define expectation of a random matrix as $E(\mathbf{Y}) = ((E(Y_{ij})))_{m \times n}$.



Dispersion

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Expectation

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Definition

The dispersion matrix or the variance-covariance matrix is

$$D(\mathbf{X}) = ((Cov(X_i, X_j)))_{n \times n} = E[(\mathbf{X} - E(\mathbf{X}))(\mathbf{X} - E(\mathbf{X}))^T] = Cov(\mathbf{X}, \mathbf{X})$$

NOTE:

$$(1) Cov(\mathbf{U}_p, \mathbf{V}_q) = ((Cov(U_i, V_j)))_{p \times q}$$

$$(2) E(\mathbf{X} + \mathbf{b}) = E(\mathbf{X}) + \mathbf{b}$$

$$(3) D(\mathbf{X} + \mathbf{b}) = D(\mathbf{X})$$

$$(4) Cov(\mathbf{X} + \mathbf{b}, \mathbf{Y} + \mathbf{c}) = Cov(\mathbf{X}, \mathbf{Y})$$



Results

Multivariate Analysis

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Expectation

Dispersion

Multivariate Normal

Quadratic forms

T-Statistic

Let $\mathbf{X}$ be a random vector with n -components such that $E(\mathbf{X}) = \mu$ and $D(\mathbf{X}) = \Sigma$ then

1 $E(l^T \mathbf{X}) = l^T \mu$, where $l \in \mathbb{R}^n$ is a constant vector

2 $D(l^T \mathbf{X}) = l^T \Sigma l$

3 $E(\mathbf{A}\mathbf{X}) = \mathbf{A}\mu$, where $\mathbf{A} \in \mathbb{R}^{p \times n}$ is a constant matrix

4 $D(\mathbf{A}\mathbf{X}) = \mathbf{A}\Sigma\mathbf{A}^T$ and $Cov(\mathbf{A}\mathbf{X}, \mathbf{B}\mathbf{X}) = \mathbf{A}\Sigma\mathbf{B}^T$

5 If $Cov(\mathbf{U}_p, \mathbf{V}_q) = \Gamma$ then $Cov(\mathbf{A}\mathbf{U}, \mathbf{B}\mathbf{V}) = \mathbf{A}\Gamma\mathbf{B}^T$



Exercise

Multivariate Analysis

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Let $\mathbf{X}$ be a random vector with n -components such that $E(\mathbf{X}) = \mu$ and $D(\mathbf{X}) = \Sigma$ then

Note1 [3]. It will imply [1].

Note2 [5]. It will imply [2] and [4].

Note3 $D(\mathbf{X})$ is a p.s.d. matrix.



Theorems

Multivariate Analysis

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Expectation

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Multivariate
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T-Statistic

Theorem 1

Let $\mathbf{X}$ be a random vector with n -components such that $E(\mathbf{X}) = \mu$ and $D(\mathbf{X}) = \Sigma$. Show that $E(\mathbf{X}^T A \mathbf{X}) = \text{trace}(\Sigma A) + \mu^T A \mu$



Theorems

Multivariate Analysis

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Theorem 2

Let $\mathbf{X}$ be a random vector with n -components such that $E(\mathbf{X}) = \mu$ and $D(\mathbf{X}) = \Sigma$ then $P((\mathbf{X} - \mu) \in \mathcal{C}(\Sigma)) = 1$.



Definition

Multivariate Analysis

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Multivariate Normal

A random vector $\mathbf{X}$ is said to follow multivariate normal $N(\boldsymbol{\mu}, \Sigma)$ if it has a density

$$f(\mathbf{x}) = \frac{\exp\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\}}{(\sqrt{2\pi})^n \sqrt{|\Sigma|}}$$

for some $\boldsymbol{\mu} \in \mathbf{R}^n$ and p.s.d. Σ

1 If $\mathbf{X} \sim N(\boldsymbol{\mu}, \Sigma)$ then $A\mathbf{X} \sim N(A\boldsymbol{\mu}, A\Sigma A^T)$



Defination

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χ^2 -distribution

If $\mathbf{X} \sim N(\boldsymbol{\mu}, \mathbf{I}_n)$ then $\mathbf{X}^T \mathbf{X}$ is said to follow Chi-squared distribution with degrees of freedom (d.f.) n and non-centrality parameter (n.c.p) $\boldsymbol{\mu}^T \boldsymbol{\mu}$.

Note: If $\mathbf{X} \sim N(\boldsymbol{\mu}, \mathbf{I}_n)$, show that $E(\mathbf{X}^T \mathbf{X}) = n + \boldsymbol{\mu}^T \boldsymbol{\mu}$



Theorem

Multivariate Analysis

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Expectation

Dispersion

Multivariate
Normal

Quadratic forms

T-Statistic

χ^2 -distribution

If $\mathbf{X} \sim N(\boldsymbol{\mu}, \mathbf{I}_n)$ then $\mathbf{X}^T \mathbf{A} \mathbf{X}$ has Chi-squared distribution iff $\mathbf{A}$ is idempotent. Moreover
 $\mathbf{X}^T \mathbf{A} \mathbf{X} \sim \chi^2_{df=Rank(\mathbf{A}), ncp=\boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}}$

Corollary : If $\mathbf{A}$ and $\mathbf{A}_1$ are symmetric and idempotent matrices such that $\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2$ and $\mathbf{A}_2$ is a p.s.d. matrix then $\mathbf{X}^T \mathbf{A}_1 \mathbf{X}$ and $\mathbf{X}^T \mathbf{A}_2 \mathbf{X}$ are independently distributed.



Independence

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Expectation

Dispersion

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Normal

Quadratic forms

T-Statistic

Theorem

Let $\mathbf{X} \sim N(\boldsymbol{\mu}, \mathbf{I}_n)$ and A is symmetric and $CA = \mathbf{0}$ then $\mathbf{X}^T A \mathbf{X}$ and $C\mathbf{X}$ are independently distributed.



Theorem

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Cochran's Theorem

Let $\mathbf{X} \sim N(\boldsymbol{\mu}, \mathbf{I}_n)$ and $\mathbf{X}^T \mathbf{A} \mathbf{X} \equiv \sum_{i=1}^k \mathbf{X}^T \mathbf{A}_i \mathbf{X}$ where $\mathbf{A}_i$ s are symmetric and $\mathbf{A}$ is an idempotent matrix. Then $\mathbf{X}^T \mathbf{A}_i \mathbf{X} \sim \chi_{\text{Rank}(\mathbf{A}_i)}^2, \boldsymbol{\mu}^T \mathbf{A}_i \boldsymbol{\mu}$ and they are independent.



T-statistic

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Normal

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T-Statistic

Let $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $S^2 = \sum_{i=1}^n (X_i - \bar{X})^2$

- 1 Find the distribution of $\bar{X}$ and S^2 .
- 2 Show that they are independently distributed
- 3 Construct t-statistic from it .
- 4 Construct F-statistic from it.



References

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Expectation

Dispersion

Multivariate
Normal

Quadratic forms

T-Statistic

- 1 (APPENDIX C1) Introduction to Linear Regression Analysis, 5th Edition By Douglas C. Montgomery, Elizabeth A. Peck, G. Geoffrey Vining
- 2 . [CHAPTER 3] Linear Models : An Integrated Approach By Debasis Sengupta, Sreenivasa Rao Jammalamadaka



SLR

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SLR

Least square

Estimation

Prediction

Regression Analysis Simple Linear Regression

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Simple linear regression

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Estimation

Prediction

- Consider a data set $D = \{(x_i, y_i) | x_i \in \mathbb{R}, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$
- x_i s are non stochastic
- y_i s are stochastic and realized values of random variable Y_i s
- ϵ_i s are iid $N(0, \sigma^2)$ random variables
- Regression parameter (β_0, β_1) and error parameter σ^2 are unknown

Problem statement

Considering the linear model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \forall i = 1, 2, \dots, n$$

we want to estimate (β_0, β_1) which will minimize least squared condition.



Least square estimate

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Estimation

Prediction

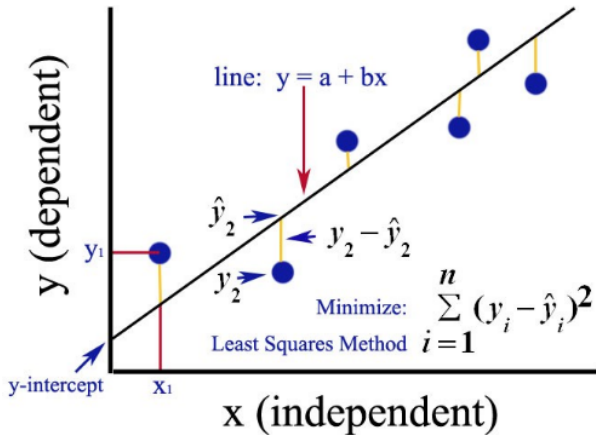


Figure: Least square



Parameter estimation

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Prediction

Least Squared condition

The least squared condition to estimate the model parameters is to minimize

$$S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2. \quad (1)$$

with respect to β_0 and β_1 .

If $(\hat{\beta}_0, \hat{\beta}_1)$ minimizes $S(\beta_0, \beta_1)$ then their values can be obtained by solving the **normal equations**

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_0} = 0 \implies n\hat{\beta}_0 + \hat{\beta}_1 \sum_i x_i = \sum_i y_i \quad (2)$$

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_1} = 0 \implies \hat{\beta}_0 \sum_i x_i + \hat{\beta}_1 \sum_i x_i^2 = \sum_i y_i x_i \quad (3)$$



Prediction

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Estimation

Prediction

Estimated parameters

Defining $S_{xy} = \sum_i (y_i - \bar{y})(x_i - \bar{x})$ we have the solutions as

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Least squared prediction line

For any x such as old x_i s or some x_{new} the prediction line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$



Prediction Error

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Estimation

Prediction

The **prediction error or residual** is defined as $e_i = y_i - \hat{y}_i$ and hence the residual sum of square (SSR) is

$$\begin{aligned}\sum_i e_i^2 &= \sum_i (y_i - \hat{y}_i)^2 \\ &= \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \\ &= \sum_i [(y_i - (\bar{y} - \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i))]^2 \\ &= \sum_i [(y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})]^2 \\ &= S_{yy} + \hat{\beta}_1^2 S_{xx} - 2\hat{\beta}_1 S_{xy} \\ &= S_{yy} - \hat{\beta}_1 S_{xy} \\ &= S_{yy} - \frac{S_{xy}^2}{S_{xx}}\end{aligned}\tag{4}$$

NOTE: We estimate σ^2 by $\hat{\sigma}^2 = \frac{SSR}{n-2} \equiv MSR$, mean residual sum of square. [Explanation will be studied under Multiple Linear Regression]



Prediction Error

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Estimation

Prediction

- We estimate σ^2 by $\hat{\sigma}^2 = \frac{SSR}{n-2} \equiv MSR$, mean residual sum of square. [Explanation will be studied under Multiple Linear Regression]
- We can estimate σ^2 even without computing individual errors e_i .
- Prediction error $\mathbf{e} = (e_1, e_2, \dots, e_n)$ and prediction vector $\hat{\mathbf{y}} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$ are independently distributed.

Linear estimator

If an estimator $T(\mathbf{y})$ can be expressed as a linear combination of $\mathbf{y}$ with non random coefficients i.e. $T(\mathbf{y}) = \sum_i \alpha_i y_i$ then $T(\mathbf{y})$ is called a linear estimator.



Note

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Estimation

Prediction

1 $\hat{\beta}_1$ is a linear estimator of β_1 . and $\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$

2 $\hat{\beta}_0$ is a linear estimator of β_0 and $\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$

3 Regression line passes through $(\bar{x}, \bar{y})$

4 $SSR \sim \sigma^2 \chi_{n-2}^2$



What more?

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Estimation

Prediction

- Can we test for $H_0 : \beta_0 = b_0$ vs $H_1 : \beta_0 \neq b_0$?
- Can we test for $H_0 : \beta_1 = b_1$ vs $H_1 : \beta_1 \neq b_1$?
- Can we test for $H_0 : \sigma^2 = \sigma_0^2$ vs $H_1 : \sigma^2 \neq \sigma_0^2$?
- Can we have a prediction interval for $\hat{y}$?

YES !!!! Now we can...



When regressor is stochastic

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Estimation

Prediction

Bivariate normal($\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho$)

If (X, Y) has joint density function $f(x, y)$ as

$$f(x, y) = \frac{e^{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_x}{\sigma_x} \right)^2 + \left(\frac{y-\mu_y}{\sigma_y} \right)^2 - 2\rho \left(\frac{x-\mu_x}{\sigma_x} \right) \left(\frac{y-\mu_y}{\sigma_y} \right) \right]}}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}$$

then (X, Y) is said to follow Bivariate normal.

- If (X, Y) follow Bivariate normal($\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho$) then show that $Y|X = x$ follows $N(\mu_y + \rho \frac{\sigma_y}{\sigma_x}(x - \mu_x), (1 - \rho^2)\sigma_y^2)$
- Perform a large and small sample test for $H_0 : \rho = 0$ Vs $H_1 : \rho \neq 0$.
- Perform a large sample test for $H_0 : \rho = \rho_0$ Vs $H_1 : \rho \neq \rho_0$.



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Prediction





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Best estimator

ANOVA

Regression Analysis Multiple Linear Regression

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Multiple linear regression

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- Consider a data set $D = \{(\mathbf{x}_i, y_i) | \mathbf{x}_i \in \mathbb{R}^{k+1}, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$
- $\mathbf{x}_i$ s are non stochastic
- y_i s are stochastic and realized values of random variable Y_i s
- ϵ_i s are iid $N(0, \sigma^2)$ random variables
- Regression parameter $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \dots, \beta_k)^T$ is unknown
- Error parameter σ^2 is unknown

Problem statement

Considering the linear model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \epsilon_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i, \quad \forall i = 1, 2, \dots, n$$

we want to estimate $\boldsymbol{\beta}$ which will minimize least squared condition.



Matrix representation

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For n such observation we use matrix notation to represent it as follows,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad (1)$$

where, $\mathbf{Y} = (y_1, y_2, \dots, y_n)^T$, $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \dots, \beta_k)^T$,

$\mathbf{X} = (\mathbf{1}, \mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_k) = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)^T$ and $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2, \dots, \epsilon_n)^T$.

Hence, there are $k + 2$ unknown model parameters, $\boldsymbol{\beta}$ and $\sigma^2 > 0$

$$\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n) \text{ as } \boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n) \quad (2)$$

Problem statement

The least square condition to be minimized to estimate $\boldsymbol{\beta}, \sigma^2$ is

$$S(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{Y}^T \mathbf{Y} - 2\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{Y} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta} \quad (3)$$



Least square

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Estimation

If $\hat{\beta}$ minimizes the least square condition then it satisfies the normal equations

$$\begin{aligned}\frac{\partial S(\beta)}{\partial \beta} \Big|_{\beta=\hat{\beta}} &= \mathbf{0} \\ \Rightarrow -2\mathbf{X}^T \mathbf{Y} + 2\mathbf{X}^T \mathbf{X} \hat{\beta} &= \mathbf{0} \\ \Rightarrow \hat{\beta} &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}\end{aligned}\tag{4}$$



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Prediction

Fitted regression for the used data is

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = P_{\mathbf{X}}\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 P_{\mathbf{X}})$$

where $P_{\mathbf{X}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is the orthogonal projection matrix of the column space of $\mathbf{X}$ i.e. $\mathcal{C}(\mathbf{X})$. It means $\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}^T\mathbf{X})$.



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Estimated Error

Hence the estimated error in prediction

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}_n - P_{\mathbf{X}})\mathbf{y} \sim N(\mathbf{0}, \sigma^2(\mathbf{I}_n - P_{\mathbf{X}}))$$

where $\mathbf{e} \in \mathcal{C}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^T \mathbf{X})^\perp$.

Note: Hence $\hat{\mathbf{y}}$ and $\mathbf{e}$ are uncorrelated and they are independently distributed when $\epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.



Definition

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Linear unbiased estimator (LUE)

A linear estimator $\mathbf{l}^T \mathbf{y} = \sum_i l_i y_i$ is said to be linear unbiased estimator (LUE) of $\mathbf{p}^T \boldsymbol{\beta}$ if $E(\mathbf{l}^T \mathbf{y}) = \mathbf{p}^T \boldsymbol{\beta}$ for all $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$.

Linear zero function (LZF)

A linear estimator $\mathbf{l}^T \mathbf{y} = \sum_i l_i y_i$ is said to be linear zero function (LZF) if $E(\mathbf{l}^T \mathbf{y}) = 0$ for all $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$



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Estimability

A linear parametric function $\mathbf{p}^T \boldsymbol{\beta}$ is said to be **estimable** if it has a LUE i.e. there exists $\mathbf{l}^T \mathbf{y}$ such that $E(\mathbf{l}^T \mathbf{y}) = \mathbf{p}^T \boldsymbol{\beta}$ for all $\boldsymbol{\beta} \in \mathbb{R}^{k+1}$.

BLUE

The best linear unbiased estimator (BLUE) of a linear parametric function $\mathbf{p}^T \boldsymbol{\beta}$ is a LUE with minimum variance.



Best estimator

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Theorem

A linear function is BLUE of its expectation iff it is uncorrelated with all linear zero functions (LZF).

Corollary

If $\mathbf{l}^T \mathbf{y}$ is an LUE of $\mathbf{p}^T \boldsymbol{\beta}$ then $\mathbf{l}^T P_{\mathbf{X}} \mathbf{y}$ is the BLUE of $\mathbf{p}^T \boldsymbol{\beta}$

Note: If you know a LUE of a parametric function then you can get the BLUE out of it.



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To test $H_0 : \beta_R = (\beta_1, \dots, \beta_k) = \mathbf{0}$ vs $H_1 : (\beta_1, \dots, \beta_k) \neq \mathbf{0}$ we perform the ANOVA as

$$SST = SSModel + SSRes$$

- $SST = \mathbf{Y}^T (\mathbf{I}_n - \frac{1}{n} \mathbf{1} \mathbf{1}^T) \mathbf{Y} \sim \sigma^2 \chi^2(df = n - 1, ncp = \lambda/\sigma^2)$
- $SSModel = \mathbf{Y}^T (P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T) \mathbf{Y} \sim \sigma^2 \chi^2(df = k, ncp = \lambda/\sigma^2)$
- $SSRes = \mathbf{Y}^T (\mathbf{I}_n - P_X) \mathbf{Y} \sim \sigma^2 \chi^2(df = n - k - 1, ncp = 0)$

Under $H_0 : \lambda = 0$, otherwise $\lambda = \beta_R^T \mathbf{X}_c^T \mathbf{X}_c \beta_R$ where $\mathbf{X}_c$ is the centred regressor variables for β_R . Hence, by Cochran's theorem we have

$$\frac{SSModel/k}{SSRes/(n - k - 1)} \sim F_{k, (n-k-1), ncp=\lambda}$$

It is a right tailed test because $\lambda = 0$ under H_0 .



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Polynomial
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Regression Analysis Polynomial Regression

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Multiple linear regression

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- Consider a data set $D = \{(x_i, y_i) | x_i \in \mathbb{R}^1, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$
- x_i s are non stochastic
- y_i s are stochastic and realized values of random variable Y_i s
- ϵ_i s are iid $N(0, \sigma^2)$ random variables
- Regression parameter $\beta = (\beta_0, \beta_1, \beta_2, \dots, \beta_k)^T$ is unknown
- Error parameter σ^2 is unknown

Problem statement

Considering the linear model

$$y_i = \beta_0 + \beta_1 x_i^1 + \beta_2 x_i^2 + \dots + \beta_k x_i^k + \epsilon_i, \quad \forall i = 1, 2, \dots, n$$

we want to estimate β which will minimize least squared condition.



Polynomial regression

Polynomial Regression

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Orthogonal PR

- We can extend the idea of multiple linear regression to polynomial regression.
- In polynomial regression we consider **higher degrees of the components x but it is linear in parameters.**
- Hence, it is a linear model too. For example,

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon \text{ for single regressor}$$

$$y = \beta_0 + \beta_1 x_1 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \epsilon \text{ for multiple regressors}$$



Matrix representation

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In general for k-degree polynomial for single regressor , we can write in matrix notation

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon \quad (1)$$

where, $\mathbf{Y} = (y_1, y_2, \dots, y_n)^T$, $\beta = (\beta_0, \beta_1, \beta_2, \dots, \beta_k)^T$, $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)^T$ with $\mathbf{x}_i = (1, x_i, x_i^2, \dots, x_i^k)^T$ and $\epsilon = (\epsilon_1, \epsilon_2, \dots, \epsilon_n)^T \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

Hence there are $k + 2$ unknown model parameters, $\beta = (\beta_0, \beta_1, \beta_2, \dots, \beta_k)^T$ and $\sigma^2 > 0$.

Least square solution

When $\mathbf{Y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n)$ the least square estimate of β will be

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.$$



Least square

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Prediction

Fitted regression for the used data is

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = P_{\mathbf{X}}\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 P_{\mathbf{X}})$$

where $P_{\mathbf{X}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is the orthogonal projection matrix of the column space of $\mathbf{X}$ i.e. $\mathcal{C}(\mathbf{X})$. It means $\hat{\mathbf{y}} \in \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}^T\mathbf{X})$.



Least square

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Estimated Error

Hence the estimated error in prediction

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}_n - P_{\mathbf{X}})\mathbf{y} \sim N(\mathbf{0}, \sigma^2(\mathbf{I}_n - P_{\mathbf{X}}))$$

where $\mathbf{e} \in \mathcal{C}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^T \mathbf{X})^\perp$.

Note: Hence $\hat{\mathbf{y}}$ and $\mathbf{e}$ are uncorrelated and they are independently distributed when $\epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.



Some Remarks

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- **Finding order of the model:** For that we can go by either (a) forward selection or (b) backward elimination.
- **Extrapolation:** Beyond the range of the data the prediction may be more erroneous.
- **Ill-conditioning:** The matrix $(\mathbf{X}^T \mathbf{X})$ may be computationally singular, specially when the magnitude of the regressor is closed to zero.
- **Hierarchy:** A model with all lower order terms of the highest degree is called hierarchical model. But regression model need not be so. In design of experiment $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_5 x_1 x_2 + \epsilon$ might be sufficient considering only individual effect and interaction effect.



Orthogonal Polynomial

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Orthogonal PR

- For a single variable polynomial regression if we increase one more degree then we need to re-estimate all the coefficients including the new one each time.
- To overcome this problem we introduce the notion of orthogonal polynomial.

Definition

For a given set of input data $\mathbf{x} = (x_1, x_2, \dots, x_n)$ a set of polynomials $\{P_0, P_1, P_2, \dots, P_k\}$ are said to be **orthogonal polynomials** if

$$P_0(x_i) = 1 \text{ and } \sum_{i=1}^n P_j(x_i)P_k(x_i) = 0 \quad \forall j \neq k \quad (2)$$



Orthogonal Polynomial

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Model

Now the model will look as follows

$$y_i = \sum_{j=0}^k \alpha_j P_j(x_i) + \epsilon_i \quad \forall i = 1, 2, \dots, n \quad (3)$$

or denoting $\mathbf{Z} = ((P_j(x_i)))_{n \times (k+1)}$

$$\mathbf{Y} = \mathbf{Z}\boldsymbol{\alpha} + \boldsymbol{\epsilon}$$

Estimation

When $\mathbf{Y} \sim N(\mathbf{Z}\boldsymbol{\alpha}, \sigma^2 \mathbf{I}_n)$ the least square estimate of $\boldsymbol{\alpha}$ will be

$$\hat{\boldsymbol{\alpha}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}.$$



Coefficient of determination

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For a given set of data $(\mathbf{y}, \mathbf{X})$ we first fit linear model under the standard assumptions. To measure how good the model is to explain dependencies between regression and independent variables we use (a) **Coefficient of determination (R^2)**:

$$\begin{aligned} R^2 &= \frac{\text{Variation in Y explained by the model}}{\text{Total variation in Y}} \\ &= \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} = \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \\ &= \frac{SS_{Model}}{SS_{Total}} \\ &= 1 - \frac{SS_{Error}}{SS_{Total}} = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \end{aligned}$$



Coefficient of determination

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NOTE:

(1) $R^2 \in (0, 1)$

(2) R^2 is an increasing function of the number of variables.

(3) Higher value of R^2 is an indicator of the better model. So we modify it to another measure of model adequacy checking.

(b) **Adjusted R^2 :**

$$\begin{aligned} R^2_{adjusted} &= 1 - \frac{\sum_{i=1}^n e_i^2 / df}{\sum_{i=1}^n (Y_i - \bar{Y}) / df} \\ &= 1 - \frac{\sum_{i=1}^n e_i^2 / (n - k - 1)}{\sum_{i=1}^n (Y_i - \bar{Y}) / (n - 1)} < R^2 \end{aligned} \quad (4)$$

NOTE: (4) $R^2_{adjusted}$ is not an increasing function of the number of variables.



Polynomial Regression

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